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Review

A Comprehensive Review of Research Works on Cooling Methods for Solar Photovoltaic Panels

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Abstract

Solar photovoltaic (PV) power is an important force in promoting the transformation of the energy structure. An increase in PV panel temperature reduces open-circuit voltage and fill factor, thereby increasing the recombination of internal charge carriers and leading to a decrease in the output power of PV systems. When PV panels operate in the environment, high solar intensity may rapidly heat PV panels to very high temperature levels, converting a significant proportion of solar energy into waste heat. Waste heat further reduces the efficiency of PV panels. Therefore, effective cooling methods are important in improving the electrical performance and reliability of PV systems. For different types of PV panel cooling methods, many research works have been conducted. Aiming at providing a relatively valuable reference for future work on PV panel cooling methods, this paper presents a comprehensive review of existing research on cooling methods for PV panels. Relevant issues of eight types of PV panel cooling methods are introduced, including working principles, typical research, advantages, disadvantages, existing problems, and future research directions.

Keywords: solar energy; PV cell; PV panel cooling; solar PV power; renewable energy utilization

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1. Introduction

Solar energy is important in energy structure adjustment and energy sustainability, and it also has significant commercial application value due to its abundant sources, flexible application methods, and low pollution of the environment [1,2]. Solar energy utilization can also serve sustainable communities [3–5], which can promote the sustainable development of society. Many solar energy utilization types have been developed, for instance, solar thermal power [6–8], solar photovoltaic (PV) power [9,10], solar-powered hydrogen generation [11–13], solar-powered desalination [14], and solar heat supply [15]. Among them, PV power is the most common solar-utilizing form. With the development of PV power technologies, the global installed capacity of PV power has been increasing year by year [16] (see Figure 1).

A PV panel is normally composed of five layers. The first layer is glass, the second is ethylene-vinyl acetate (EVA) material, the third is a PV cell, the fourth is a lower EVA layer, and the fifth is the backboard made of various materials (e.g., Tedlar). Additionally, PV panels are usually equipped with metal outer frames (e.g., aluminum alloy).

Relevant research shows that the theoretical photo-electric efficiency of single-crystal silicon cells under standard test conditions can be 33.7%, and currently, the efficiency range of commercial high-efficiency crystalline silicon PV panels is 12–18%. An increase

in PV panel temperature reduces open-circuit voltage and fill factor, thereby increasing the recombination of internal charge carriers and leading to a decrease in output power. When the PV panel temperature increases by 1.0 °C, the photo-electric efficiency decreases by 0.4–0.65% [17]. Figure 2 presents the influence of temperature on the voltage–current behavior of a single-crystal silicon PV panel. Many factors (e.g., wind speed and direction, solar intensity, dust accumulation, and humidity) can impact PV panel temperature. When PV panels operate in the environment, high solar intensity may rapidly heat PV panels to very high temperature levels, converting a significant proportion of solar energy into waste heat. Waste heat further reduces PV panel efficiency. Therefore, effective cooling methods are important in improving the electrical performance and reliability of PV systems.

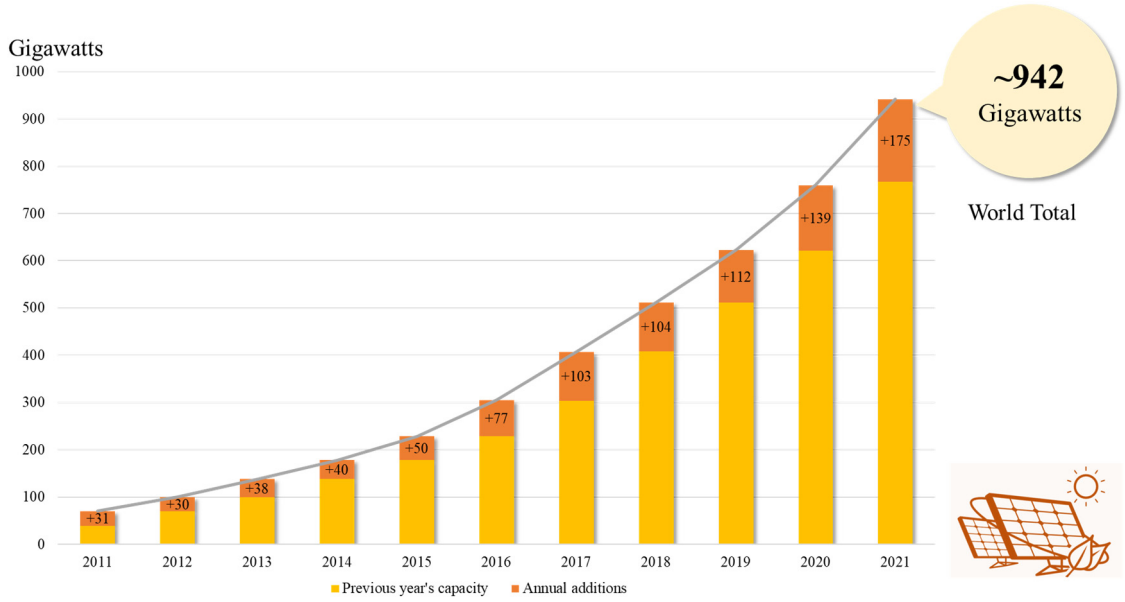


Figure 1. Global installed PV capacity during 2011–2021.

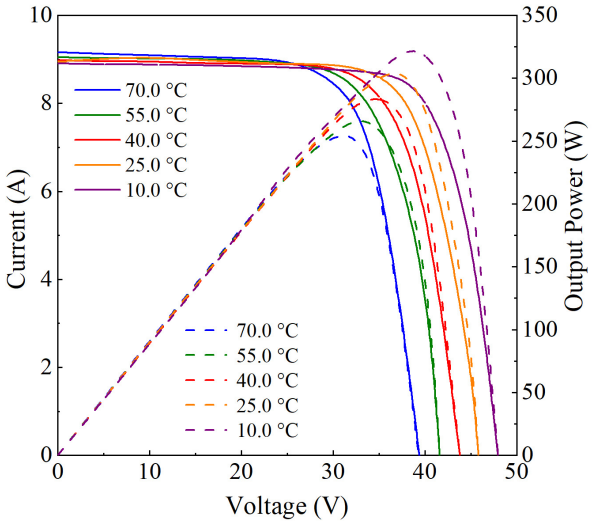


Figure 2. Influence of temperature on the voltage–current performance of a single-crystal silicon PV panel.

Many cooling technologies can be utilized to reduce PV panel temperature. In general, cooling technologies for PV panels mainly include two types, namely, active and passive cooling technologies [18]. Active cooling technologies usually utilize forced flows driven by various devices (e.g., pump and fan), and passive cooling is normally achieved by using

natural circulation, heat conduction, or radiation heat transfer. The cooling effects of active cooling methods are better, and the typical advantage of the passive cooling method is that no external power is consumed.

Many studies on cooling methods for solar PV panels have been conducted by researchers, but so far, there is still a lack of a comprehensive review of studies on cooling approaches for solar PV panels. Compared with previous works, the major contribution of this paper is providing a comprehensive review of research on cooling methods for PV panels. According to different cooling types, typical studies on air, liquid, heat pipe, phase-change-material (PCM)-based, spectrum-based, evaporative, thermo-electric, and composite cooling methods for PV panels are reviewed. Principles and applications of these cooling methods reported by the typical literature will be introduced to provide a reference for future research on cooling technologies for PV systems.

2. Air Cooling

Air is lightweight and easy to transport. Air medium cooling systems can have low cost and simple maintenance work and can reduce the related risks of freezing or leakage. Therefore, it is common to use air to cool PV panels. According to the flow types of air, air cooling methods for PV panels can be divided into natural circulation and forced circulation air cooling methods.

2.1. Natural Circulation Air Cooling

The natural circulation air cooling method belongs to the passive cooling technology. It takes heat away from PV panels through natural circulation flow on the front or back of the PV panels, reducing the PV panel temperature. An effective method for the natural circulation air cooling of PV panels is to add heat sinks on the back of the PV panels to further enhance the cooling effect [19]. The heat sink is normally a thermal conductor (e.g., fin) which can absorb heat from PV panels and dissipate it to the surrounding environment [20]. Appropriately increasing the heat transfer area of fins or adjusting the fin layout can enhance the PV panel cooling effects [21].

Many studies on natural circulation air cooling for PV panels through heat sinks have been conducted. Bayrak et al. [22] analyzed the influences of fin parameters on the temperature and electrical power of PV panels under the natural convection condition through experiments. The experimental devices in the study are presented in Figure 3. They found that the PV panel using fins with 7.0 cm $\times$ 20.0 cm dimensions had the best cooling performance, with the output power being 9.4 W higher than that of the PV panel with no fins. Another similar experimental study was conducted by Selimefendigil et al. [23], who cooled PV panels by porous fins. The results indicated that the use of porous fins brought an increase of 7.26 W in output power. Another typical fin cooling method for PV panels was studied by Hernandez-Perez et al. [24], and the diagram is illustrated in Figure 4.

Kong et al. [25] conducted a performance evaluation on a roof-type building-attached photovoltaic (BAPV) system with double-channel air flow cooling through the experimental method (see Figure 5). In their study, the BAPV system was designed to have a double air flow cooling channel structure. The experimental results revealed that compared with the conventional BAPV system, the BAPV system with double-channel air flow cooling could have a PV panel temperature reduction of 6.2 °C and a 4.5% increase in average output power.

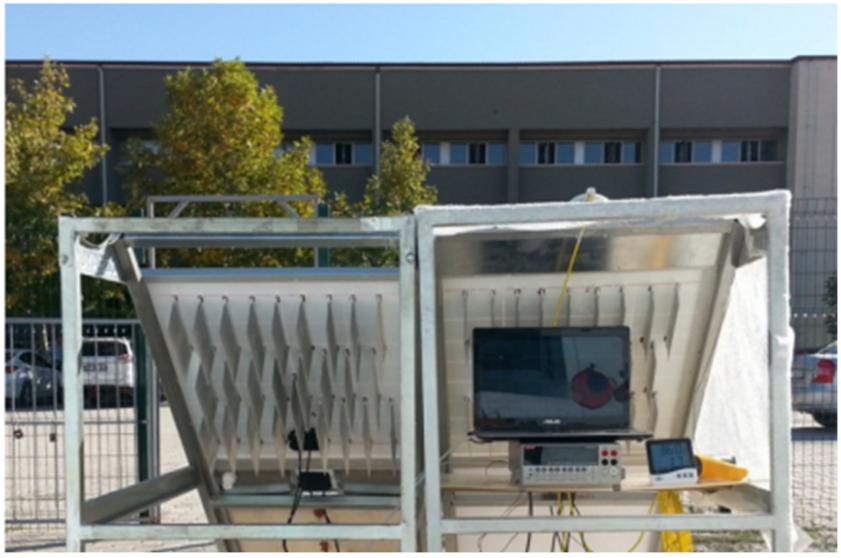


Figure 3. Experimental devices of fin-based natural circulation air cooling method for PV panels [22].

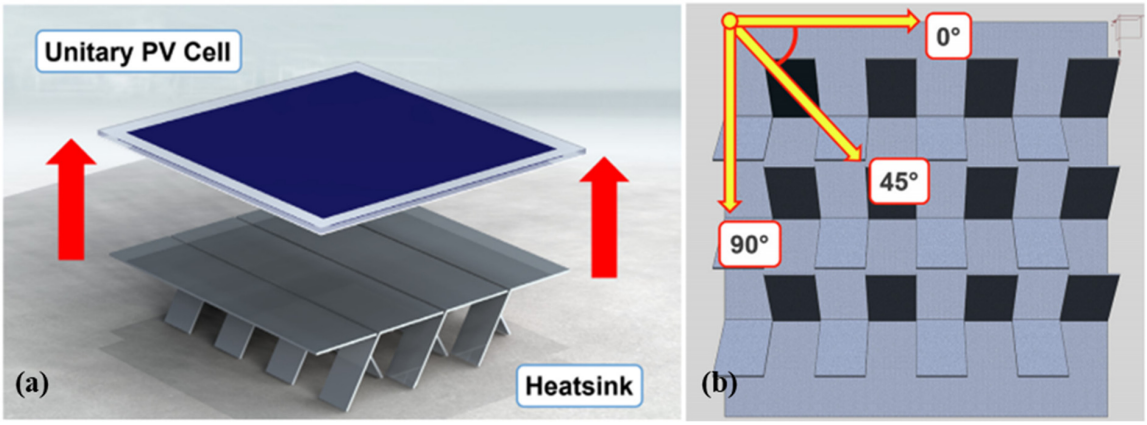


Figure 4. Diagram of the heat sink fin array proposed by Hernandez-Perez et al.: (**a**) overall structure and (**b**) explanation of fin and airflow directions [24].

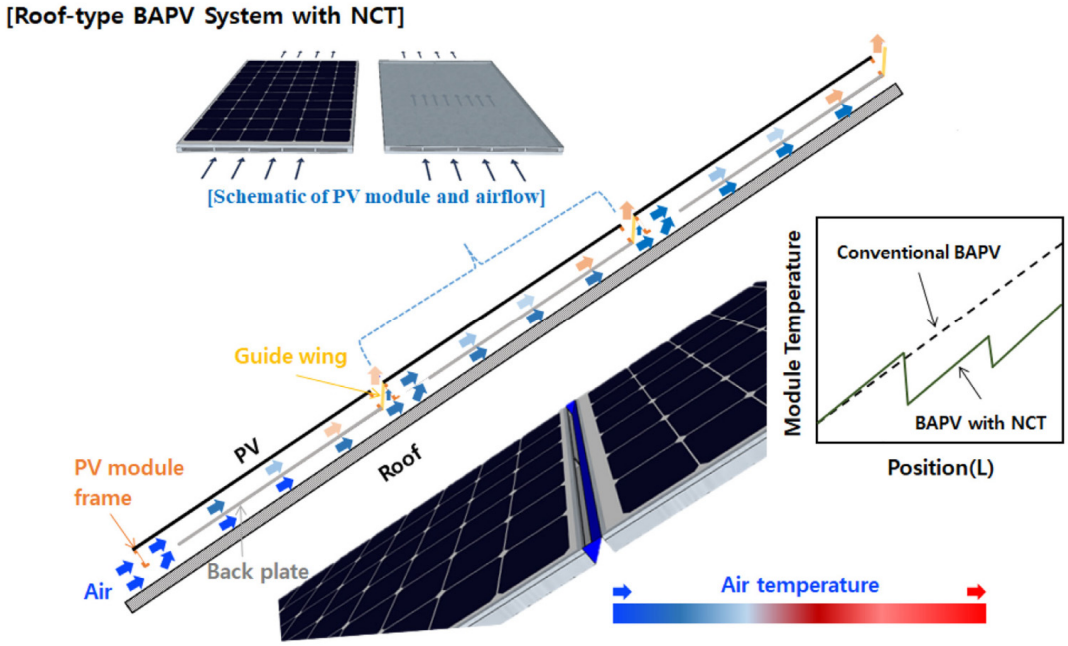


Figure 5. Diagram of a BAPV system with double-channel air flow cooling [25].

Razali et al. [26] proposed a conical fin heat sink design for PV panels. The heat sink comprised fins with different heights and directions (see Figure 6) and could effectively accelerate the heat transfer in the cooling system of PV panels. For the experimental device, the heat sink was made of aluminum alloy, and the PV panel was of the half-cut-cell–PERC type. The evaluation showed that compared with the PV panel without the heat sink, the one using the conical fin heat sink could have a temperature reduction of 12.0 °C.

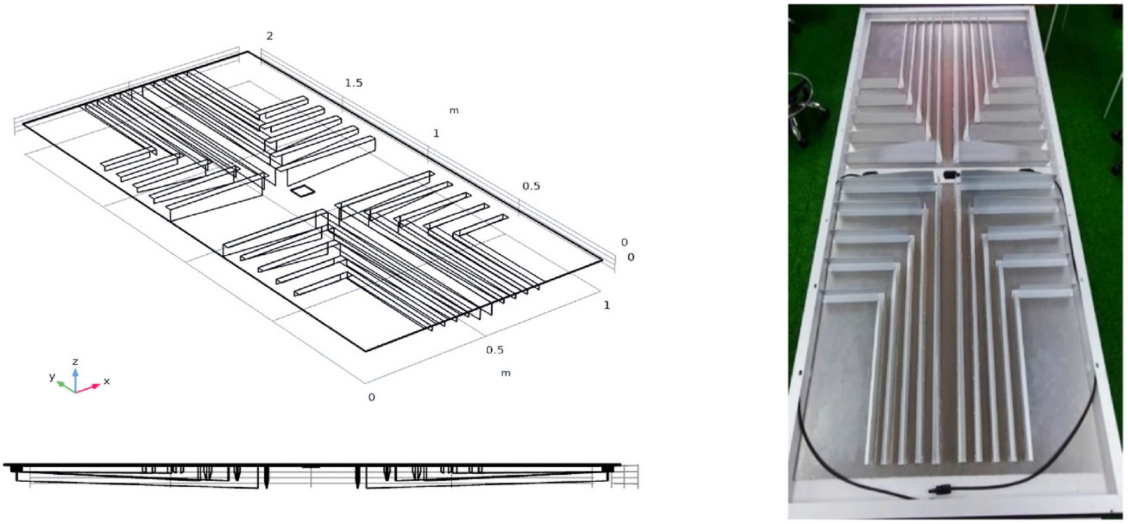


Figure 6. Diagram and experimental device of conical fin heat sink for PV panels [26].

In addition, some researchers also proposed to improve the air flow condition through the modification of the PV panel structure. For instance, Abd-Elhaby et al. [27] evaluated the feasibility of enhancing the passive cooling of PV panels by drilling holes in PV panels. The diagram of this method is illustrated in Figure 7. Relevant experiments and simulations were conducted. The results showed that when the PV panel had a hole density of 46.0 hole/m^2, the panel temperature reduction was 19.0 °C.

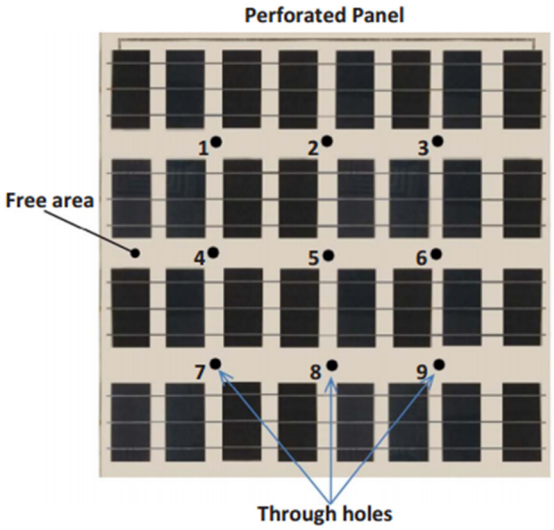


Figure 7. Passive cooling enhancement of PV panels by drilling holes in panels [27].

Marinić-Kragić et al. [28] investigated a passive cooling enhancement method for PV panels by introducing slits in PV panels. The cooling performance of PV panels with two different slit directions was investigated through the simulation method (see Figure 8). The

results revealed that by introducing slits in the PV panels, the panel temperature reduction could be up to 3.0 °C.

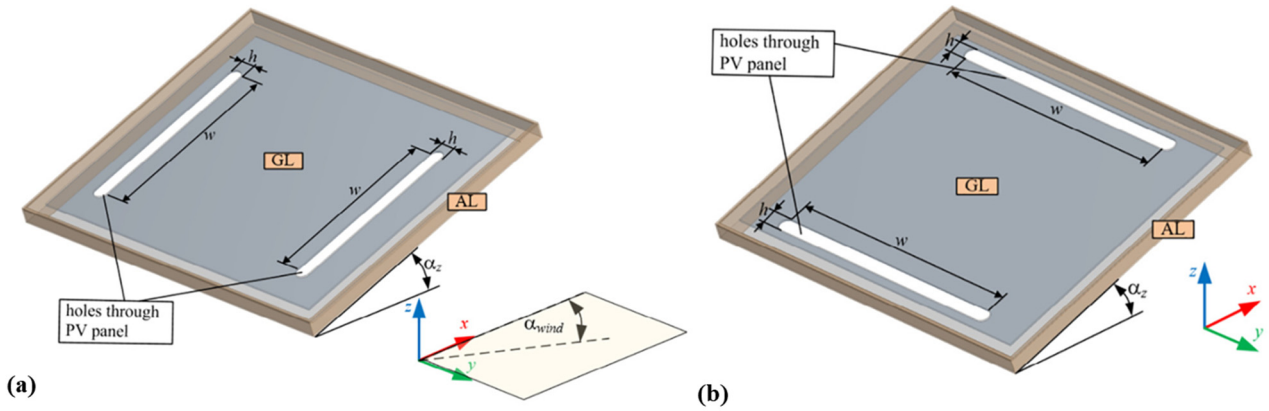


Figure 8. Slit directions in PV panels: (**a**) parallel to x-axis and (**b**) parallel to y-axis [28].

2.2. Forced Circulation Air Cooling

The forced circulation air cooling method belongs to the active cooling technologies [29]. It requires the use of mechanical devices such as fans to enhance the airflow through PV panels, thereby achieving better cooling effects [30]. This subsection will introduce some typical studies on forced circulation air cooling methods for PV panels.

Mojumder et al. [31] studied a PV system with single-channel air active cooling (see Figure 9). Some rectangular fins were installed in the air flow channel to enhance the cooling performance. The analysis results showed that when the flow rate of air and incident solar intensity were 0.14 kg/s and 700.0 W/m^2, the highest thermal and photo-electric efficiency rates of the PV system with four fins were 56.2% and 13.8%.

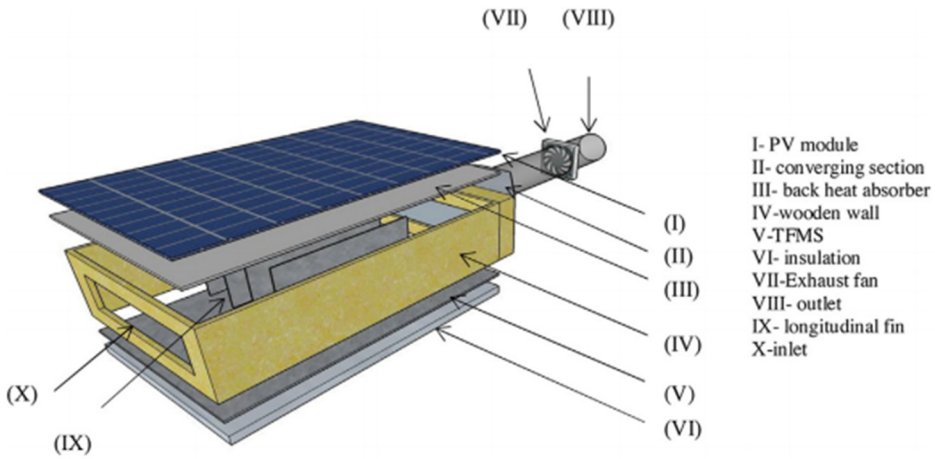


Figure 9. A PV system with single-channel air active cooling [31].

Li et al. [32] conducted various tests on a PV panel cooling method based on compressed air circulation. The circulation mainly consisted of compressor, air storage tank, regulation valve and nozzles (see Figure 10). The compressor was designed to be driven by PV panels. The valve could regulate the air flow rate to meet the demands of panel cleaning and cooling. Experimental tests of dust elimination and the cooling of PV panels were conducted. The results showed that when the blowing time was 20.0 s, the output power of the PV array increased from 567.4 W by 36.1%.

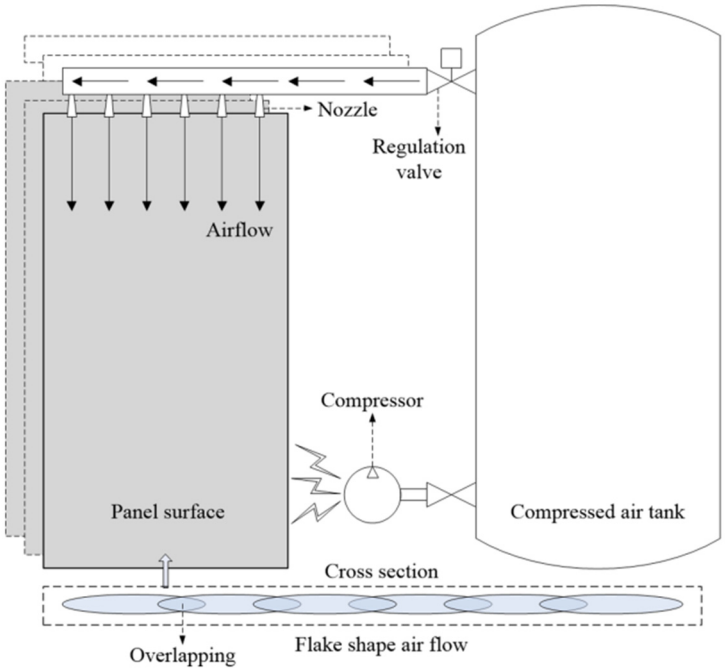


Figure 10. Diagram of a PV panel cooling system based on compressed air circulation [32].

Wang et al. [33] proposed a scenario which could improve the forced convection of roof-installed PV panels. The scenario was composed of curved eaves and vortex generators arranged on the roof. The arrangement of the vortex generators was optimized. The results indicate that by using that scenario, the forced convection heat transfer ratio on the PV panel surface could be 60.0%, and the highest panel temperature reduction was 5.9 °C.

2.3. Brief Summary of Air Cooling Methods

Besides studies on air cooling methods based solely on natural or forced circulation, some studies have presented both natural and forced air cooling methods for PV panels simultaneously [34,35], with some comparing the performance of a certain PV panel with the two types of air cooling methods.

Air cooling technology largely depends on air flow conditions, and the cooling performance is heavily affected by environmental factors. Future studies on air cooling methods for PV panels may include the following:

(a) More studies on enhancing the passive natural convection air cooling method are suggested to be conducted, mainly focusing on the optimizations of fin parameters and arrangement, the design of new air flow channels, and the new structural renovations of PV panels (e.g., drilling holes and grooving).

(b) For the forced circulation air cooling method, economic evaluations on the relationship between the power increase in PV systems and the external power consumed by the cooling system are suggested to be conducted. This is very meaningful for evaluating the practicality of the studied active air cooling methods.

3. Liquid Cooling

Water and some other liquids are coolant materials which can be used in active cooling technologies for reducing the PV panel temperature [36,37]. The liquid cooling system needs to ensure the electrical insulation between the PV cell and the cooling medium. According to the different cooling methods of the working fluid, liquid cooling methods for PV panels mainly include channel-type cooling [38,39], surface-type cooling [40,41], liquid immersion cooling, and floating cooling.

3.1. Channel-Type and Surface-Type Cooling

Channel-type cooling refers to a heat dissipation method in which the liquid circulating in the channel directly or indirectly comes into contact with PV panels, taking away heat from the panels. This subsection will introduce some typical studies on channel-type liquid cooling methods for PV panels.

Ebaid et al. [42] conducted some experiments on the cooling performance of PV panels using water and a nanofluid, and Figure 11 presents the experimental device. The cooling performance of PV panels using water, Al_2O_3/water nanofluid, and T_iO_2/water nanofluid was estimated. The results revealed that compared with uncooled PV panels, the panels using water and nanofluid cooling methods had lower temperatures. The Al_2O_3 nanofluid had better cooling capacity compared with the T_iO_2 nanofluid. By using the Al_2O_3 nanofluid with a 0.1 wt.% concentration and a flow velocity of 5000.0 mL/min, the PV panel could reach a temperature reduction of 17.0 °C.



Figure 11. An experimental device of PV panels with liquid cooling [42].

Zhang et al. [43] conducted a numerical study on a multi-hole water cooling channel for PV panels. The results demonstrated that the non-uniform distribution of holes near the water outlet had better cooling performance. The optimal cooling effect was achieved when the hole diameter was 5.0 mm. The overall efficiency of the PV module with that cooling channel was 4.2% higher compared with the conventional one when the solar intensity was 1000.0 W/m^2.

Relevant research results reveal that channel-type liquid cooling methods normally need a circulating pump. Therefore, many researchers have proposed to collect the working fluid for thermal utilization; then, the PV system can be considered a PV/T system. As this kind of PV/T system does not utilize spectral filtering technologies, it belongs to the category of traditional PV/T systems. For instance, Kianifard et al. [44] proposed a design of a PV/T system using half-pipe water cooling. Figure 12 presents the experimental device of the PV/T system which was used to verify the mathematical model of the PV/T system. A half-round pipe was used instead of a full-round pipe. A pump was utilized to drive water to flow in the channel. The analysis results indicated that the proposed cooling method could make the electric efficiency of the PV/T system 0.4–0.6% higher compared with the conventional system.

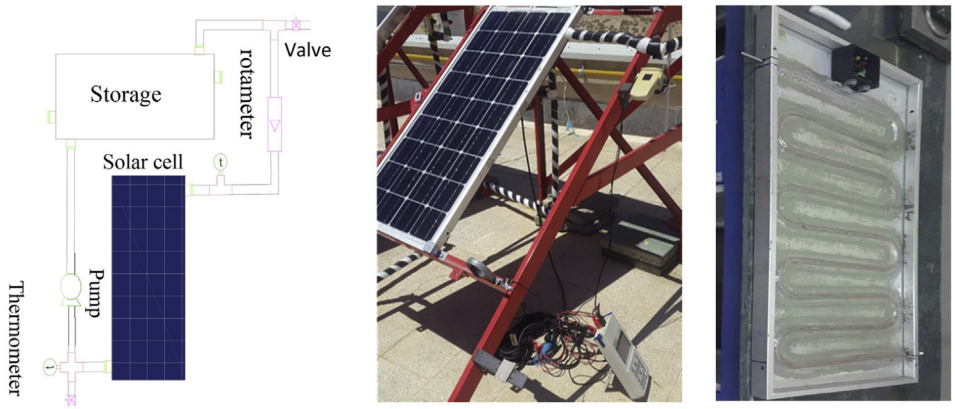


Figure 12. Experimental device of a PV/T system using half-pipe water cooling: (**a**) diagram, (**b**) physical photo, and (**c**) photo of the back of PV panel [44].

Nahar et al. [45] conducted a numerical study on a PV/T system with channel-type water cooling by using COMSOL software (V5.3). They found that when the water channel depth was 20.0 mm, a PV panel temperature reduction of 10.2 °C could be obtained.

Surface-type cooling methods normally utilize liquid working media to be sprayed onto the front or back of PV panels to achieve the effect of cooling the panels [46]. Surface-type cooling technology has advantages such as high heat transfer coefficient and non-contact thermal resistance [47]. In addition, it can exert a dust removal effect while cooling the PV panels [48]. Many relevant studies have been conducted, and some of them are introduced in this subsection.

Khan et al. [49] studied a method of utilizing a water spray to cool PV panels. For evaluating the cooling effect, an experimental device was built in Sultanpur, India. The tests were conducted under different climatic conditions. The results showed that the evaluated method could decrease the PV panel temperature from 53.0 °C to 23.0 °C.

Javidand and Moghadam [50] developed an experimental setup for PV panels with SiC/water nanofluid nozzle cooling (see Figure 13). Effects of arrays of two kinds of nozzles (single- and multi-orifice nozzles) on the operation performance of PV panels were evaluated. The results revealed that when the flow rate and concentration of the nanofluid were 0.12 kg/s and 1.1 wt.%, the multi-orifice nozzle could induce a panel temperature reduction of 6.7 °C.

Yesildal et al. [51] carried out some experiments to evaluate effects of an air-aided water spray cooling method on the performance of PV panels. Figure 14 presents the experimental setup. The results indicated that when the solar intensity was 900.0 W/m^2, the electric efficiency of the PV panel using spray cooling was 10.2%, which was higher than that of the panel without spray cooling (7.9%).

Zhao et al. [52] evaluated the performance of PV panels under spray cooling conditions by using a mathematical model. The output power and electric efficiency of PV panels under various cooling conditions were evaluated. The results indicated that spray cooling had better cooling performance for PV panels compared with common water cooling. It could bring a panel temperature reduction of 28.0 °C compared with the PV panels with air cooling under the non-solar concentration condition.

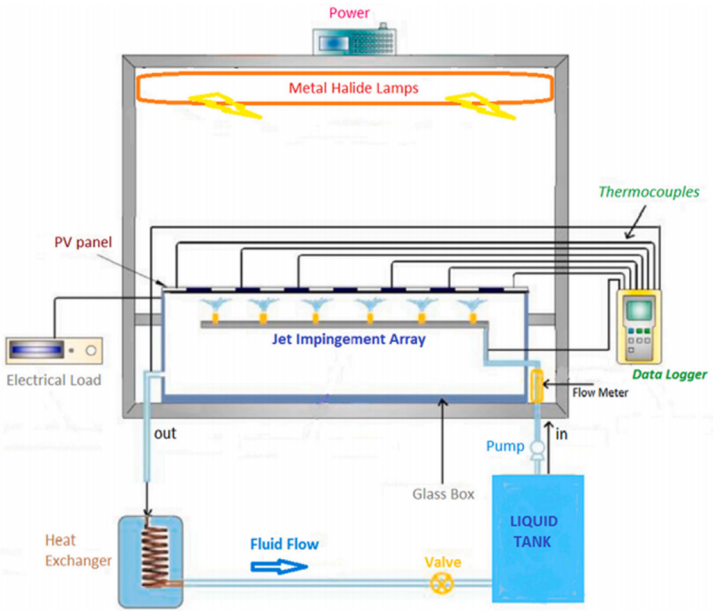


Figure 13. Diagram of an experimental nozzle cooling device for PV panels [50].

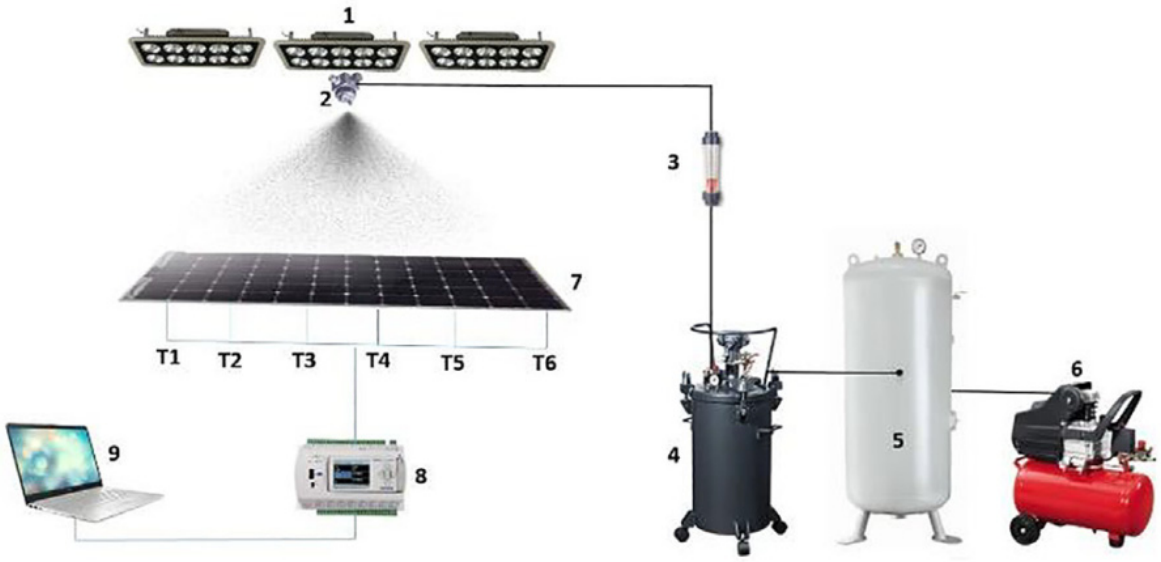


Figure 14. Diagram of an experimental device for PV panels with air-aided water mist spray cooling (1—halogen projector; 2—spray nozzle; 3–flowmeter; 4–spray liquid tank; 5–pressured air tank; 6–compressor; 7–PV panel; 8–data logger; 9–computer) [51].

3.2. Immersion and Floating Cooling

Immersing PV panels in liquid is an efficient cooling method [53]. Liquid immersion cooling refers to the cooling method of immersing PV panels in a stationary or circulating cooling medium so that the cooling medium can directly come into contact with and cool the PV panels.

Tina et al. [54] conducted experiments with a water immersion cooling method for PV panels, as shown in Figure 15. The results demonstrated that the PV panel had the highest output power when it was immersed in water at a depth of 5.0 cm. At that immersion depth, the output power increased by 19.4% compared with non-immersed PV panels.

Abdulgafar et al. [55] conducted a study on a water immersion cooling method for PV panels through the experimental method. The photo-electric efficiency of a PV panel was tested by immersing it in distilled water at various depths. The results indicated that

when the water immersion depth was 6.0 cm, the photo-electric efficiency of the PV panel increased by 11.0%.

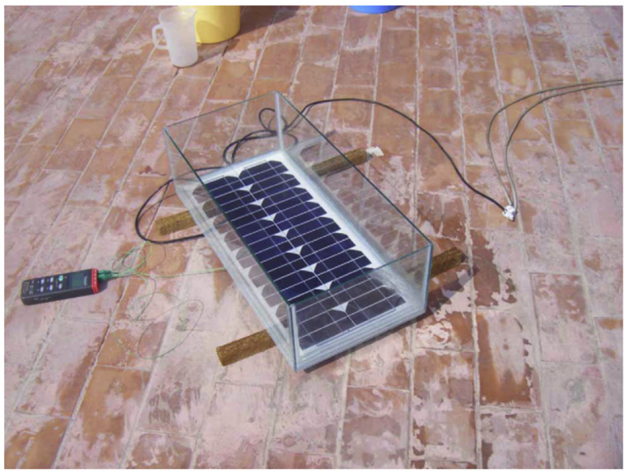


Figure 15. An experimental device for immersing PV panels [54].

Xin et al. [56] evaluated the performance of a GaInP/GaInAs/Ge PV panel with liquid immersion cooling through both the experimental and simulation methods. The liquid used for immersion was simethicone, and the immersion depth was changed from 1.0 mm to 30.0 mm. The results showed that when the immersion depth was 1.0 mm, compared with uncooled PV panels, the photo-electric efficiency of the immersed panel increased from 39.6% to 40.6%.

Elminshawy et al. [57] conducted an experimental evaluation on a floating PV system constructed in Egypt with partial water immersion cooling (see Figure 16). The results revealed that the floating PV system could have a PV panel temperature reduction of 15.1% and an output power increase of 20.76% compared with land PV systems.

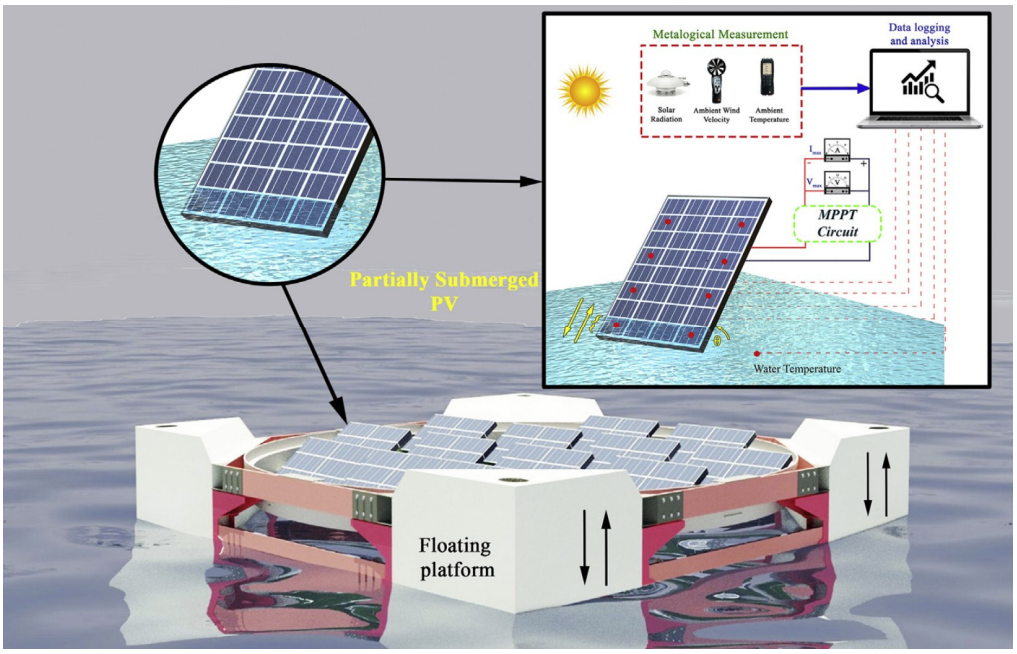


Figure 16. Diagram of an experimental setup of a floating PV system with partial water immersion cooling [57].

Floating cooling refers to the use of evaporated water vapor to cool PV panels through natural convection, which exits in floating PV systems [58]. Compared with land PV systems, floating PV systems can not only solve the PV panel cooling problem but also bring some other benefits [59]. For instance, in the economic aspect, the floating PV system can have lower O&M costs [60]. In the environmental aspect, floating PV systems can reduce water evaporation by about 70.0% [61] and limit the growth of algae, reducing water pollution. Many studies on floating PV systems have been conducted, and some of them will be presented in this subsection.

Liu et al. [62] established a finite element model to evaluate the cooling effect of water on a floating PV system. The analysis results indicated that compared with land PV systems, the PV panel temperature of the floating PV system was 3.5 °C lower, and the photo-electric efficiency was 1.58~2.0% higher.

Elminshawy et al. [63] proposed to use partial immersion angle perforating fins to enhance the cooling of a floating PV system and conducted some experiments. Figure 17 shows the experimental setup. They found that when the flow velocity of water, wind speed, and wind direction were 0.3 m/s, 5.0 m/s, and 60.0° and partial immersion angle perforating fins were used, the PV panel temperature decreased by 33.3%, leading to a 22.8% increase in photo-electric efficiency.



(a) Front view (b) Back view

Figure 17. Experimental setup of a floating PV system with partial immersion angle perforating fins [63].

Hammoumi et al. [64] conducted some experiments on a floating PV system. The experimental device is presented in Figure 18. The results demonstrated that the average temperature of the floating PV panel was 2.74 °C lower than that of the land PV panel. Additionally, the daily power generation of the floating PV system was 2.33% higher than that of land PV systems.

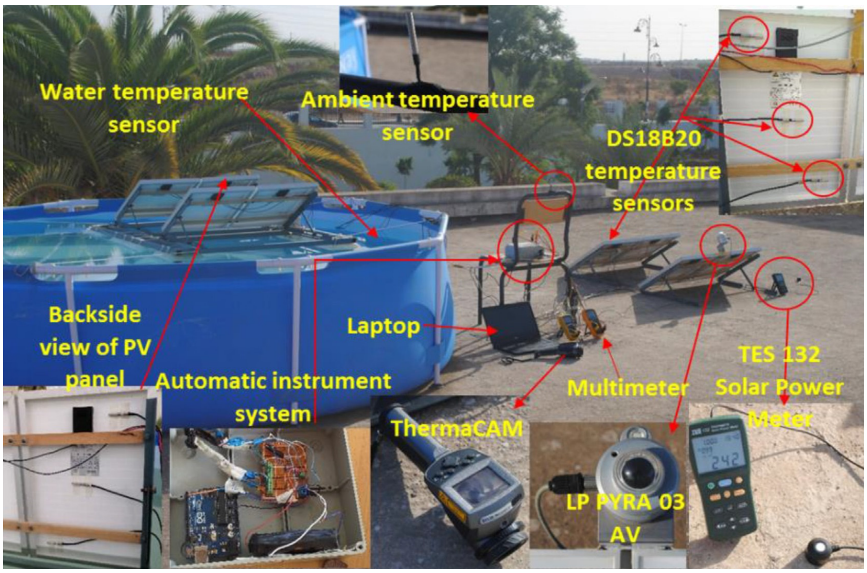


Figure 18. Experimental device of the floating PV system studied by Hammoumi et al. [64].

3.3. Brief Summary of Liquid Cooling Methods

The advantages of different kinds of liquid cooling methods have been introduced in the above. Nevertheless, these liquid cooling methods also have some problems. Potential further research contents are summarized as follows:

(a) To make forced liquid cooling methods for PV panels more feasible, further studies may include heat transfer enhancement by modifying the liquid flow channel, optimization of spray nozzle structure and arrangement, pulse jet or spray cooling, and economic feasibility analyses of cooling methods.

(b) The liquid immersion cooling performance for PV panels can be influenced by the immersion depth due to different optical absorption characteristics of different liquids. Further studies on PV panel temperature uniformity analysis and the evaluation of the immersion depth's effect on the operation performance of PV panels will be meaningful.

(c) For floating PV systems, some issues still need to be further studied, for instance, optimization of the PV panel arrangement, cooling performance evaluations of the thin-film PV panels used in floating PV systems, and economic feasibility analysis. Additionally, some issues unrelated to PV panel cooling should also be considered, such as the evaluation of the water corrosion effect on the structure and operation of PV panels and the analysis of the effect of floating PV systems on ecological footprint, water quality, and other environmental factors.

4. Heat Pipe Cooling

Heat pipe is an effective passive heat transfer element which utilizes phase change in a medium inside a fully enclosed vacuum pipe for heat transfer. It mainly consists of three parts, namely, evaporation, insulation, and condensation sections [65]. The heat transfer inside a heat pipe primarily depends on the gas–liquid phase change in the working fluid and does not require a large temperature difference between the heat source and the heat sink. It can have high temperature uniformity, high thermal conductivity, and variable heat flux without additional energy consumption [66]. Additionally, heat pipes also have advantages of low cost, high reliability, long service life, and diverse structure and can be freely designed according to the heat dissipation requirements and structural characteristics of different PV systems.

Concentrated solar photovoltaic (CPV) systems with heat pipe cooling can not only obtain effective cooling effects for PV panels but also collect some heat for thermal utilization.

When heat pipes for PV systems are designed, water, ethanol, ammonia, toluene, pentane, and other materials can be selected as the working fluid.

The currently widely used types of heat pipes include single-pipe heat pipes with a wick, loop heat pipes, two-phase closed gravity heat pipes, separated heat pipes, pulsating heat pipes, and micro-channel heat pipes [67]. PV panel cooling technologies based on heat pipes have received much attention due to their advantages such as simple structure, no energy consumption, and efficient heat dissipation [68].

Hu et al. [69] conducted some experiments on solar PV/T systems using heat pipes without a wick and with a wire mesh, respectively. Figures 19 and 20 present the diagram of the studied PV/T system and two kinds of heat pipes. The effect of the tilt angle on the thermal performance of the PV/T system was evaluated. The results demonstrated that the thermal performance of the heat pipe without a wick was sensible to the tilt angle, while that of the heat pipe with a wire mesh was not sensible to the tilt angle.

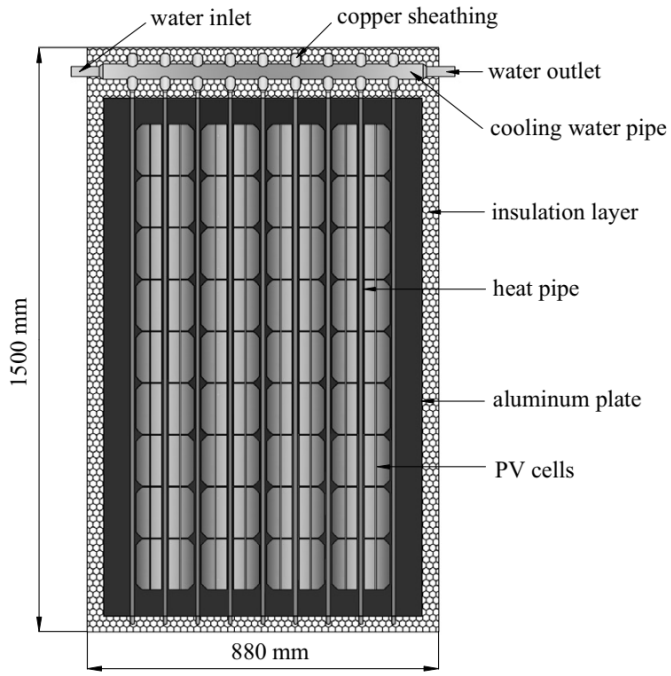


Figure 19. Diagram of a PV/T system with heat pipe cooling studied by Hu et al. [69].

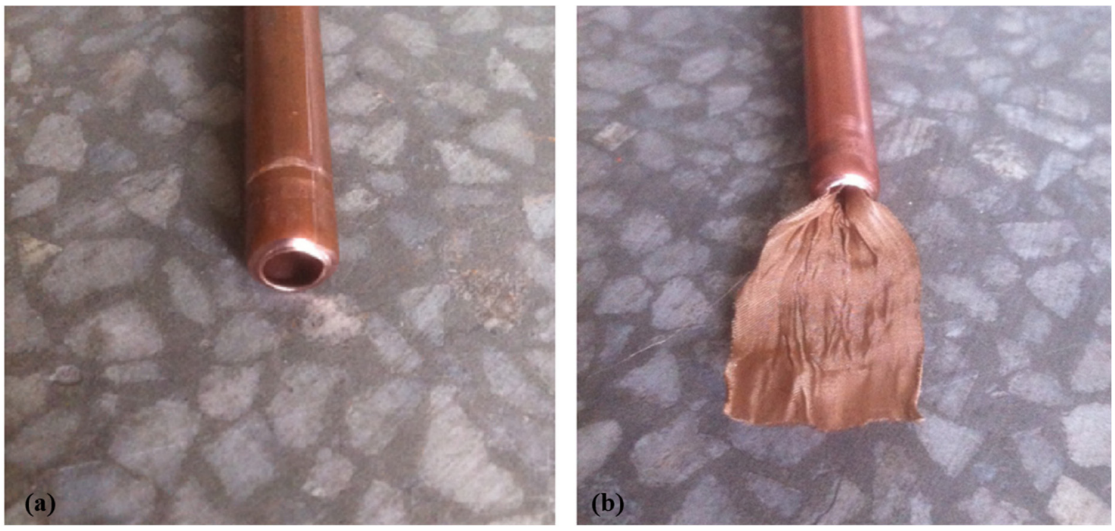


Figure 20. Photos of heat pipes without a wick (**a**) and with a wire mesh (**b**) [69].

Heat pipes with a wick can be made into single pipes or loop heat pipes. The most widely used wick structure is the copper powder sintered liquid-absorbing wick, which can make the heat transfer of the heat pipe unaffected by the direction of gravity, thus making the heat pipe more applicable [70].

Pulsating heat pipes have relatively good heat transfer capacity and thus can improve the PV panel cooling effect [71]. Alizadeh et al. [72] evaluated the cooling effect of a pulsating heat pipe on the PV panel through the simulation method (see Figure 21). The results revealed that when the solar intensity was 1000.0 W/m^2, compared with uncooled PV panels, the output power of the panel with the pulsating heat pipe increased by 18.0%.

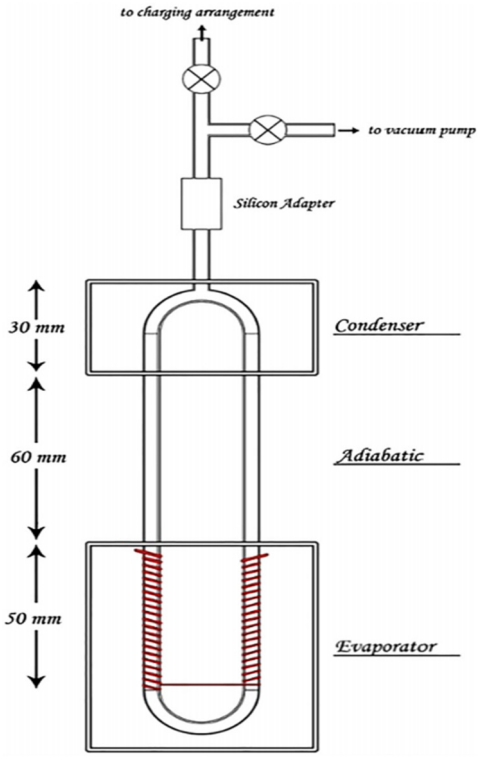


Figure 21. Diagram of a pulsating heat pipe for cooling PV panels [72].

There is a significant thermal contact resistance between traditional columnar heat pipes and PV panels, which affects the heat transfer efficiency in the evaporation section. The micro-channel heat pipes have a flat shape, which allows them to come into contact appropriately with the PV panel. Wang et al. [73] conducted an investigation on an air-cooled PV/T system with micro-channel heat pipes through the experimental method (see Figure 22). The results indicated that the PV panel temperature decreased by 22.8 °C and the electric efficiency increased by 30.9%.

Some challenges for heat pipe cooling methods for PV panels still exist [67], and relevant future research contents are suggested as follows:

(a) The complex latent heat transfer and convective heat transfer mechanisms inside some kinds of heat pipes (e.g., pulsating heat pipes) still need further exploration.

(b) Most studies on heat pipe cooling methods for PV panels mainly focus on the maximum temperature or average temperature of PV panels. Evaluations of the effect of heat pipe cooling on PV panel temperature uniformity have rarely been reported. This issue needs to be addressed in the future.

(c) Efficiency research, economic cost–benefit analyses, and environmental benefit evaluations of the long-term operation of heat pipe-cooled PV systems under actual conditions are still relatively rare and need to be conducted.

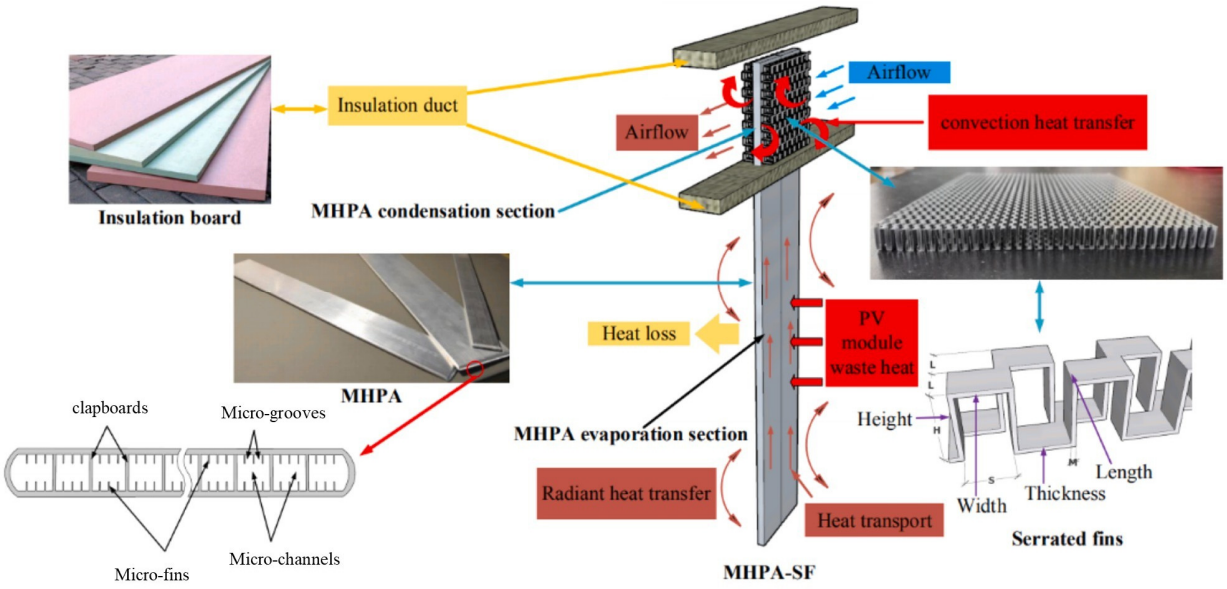


Figure 22. Workflow of an air-cooled PV/T system with micro-channel heat pipe [73].

5. PCM-Based Cooling

Different from the heat pipe cooling methods, the PCM-based cooling methods for PV panels introduced in this section refer to cooling methods which utilize solid–liquid phase change [74]. The typical PV/PCM integrated system mainly comprises a PV panel and a PCM container made of high-thermal-conduction metal [75]. Scholars have conducted extensive simulation and experimental studies on PV systems using PCM cooling.

5.1. Pure PCM-Based Cooling

For typical simulation studies on PV panel cooling based on pure PCMs, Kant et al. [76] studied the effects of convection inside melted PCMs, wind speed, and tilt angle of the PV panel on the cooling performance of a PV system (see Figure 23) with PCM cooling through simulations. The PCM was assumed to be RT35. The simulation results indicated that when both the convection and conduction heat transfers in the PCM were considered, the PV panel temperature could be reduced by 6.0 °C.

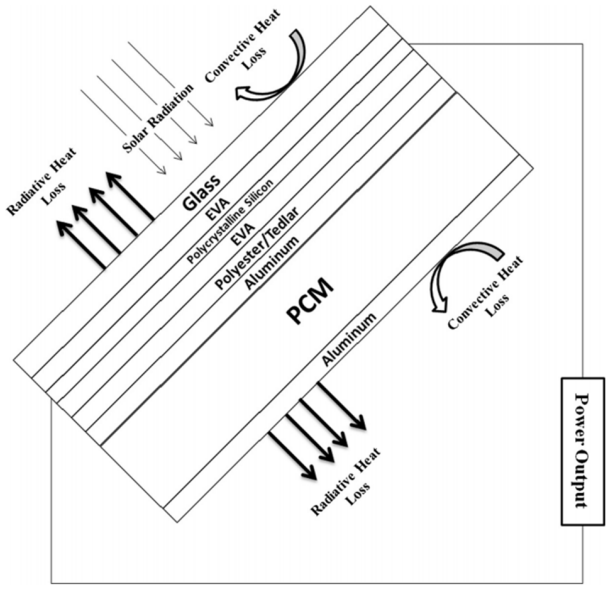


Figure 23. Diagram of a PV system using PCM cooling studied by Kant et al. [76].

For typical experimental studies, Ranawa and Nalwa [77] evaluated the cooling performance of PV panels using different multi-layer PCM cooling schemes through the experimental method. Figures 24 and 25 present the diagrams and experimental device for PV panels with multi-layer PCM cooling. OM37 and OM42 were used as the PCMs. The results revealed that compared with the PV panel with only single-layer OM42, the temperature reductions of the panels using OM37/OM37/OM42 and OM37/OM42/OM42 were 3.0 °C and 1.9 °C, respectively.

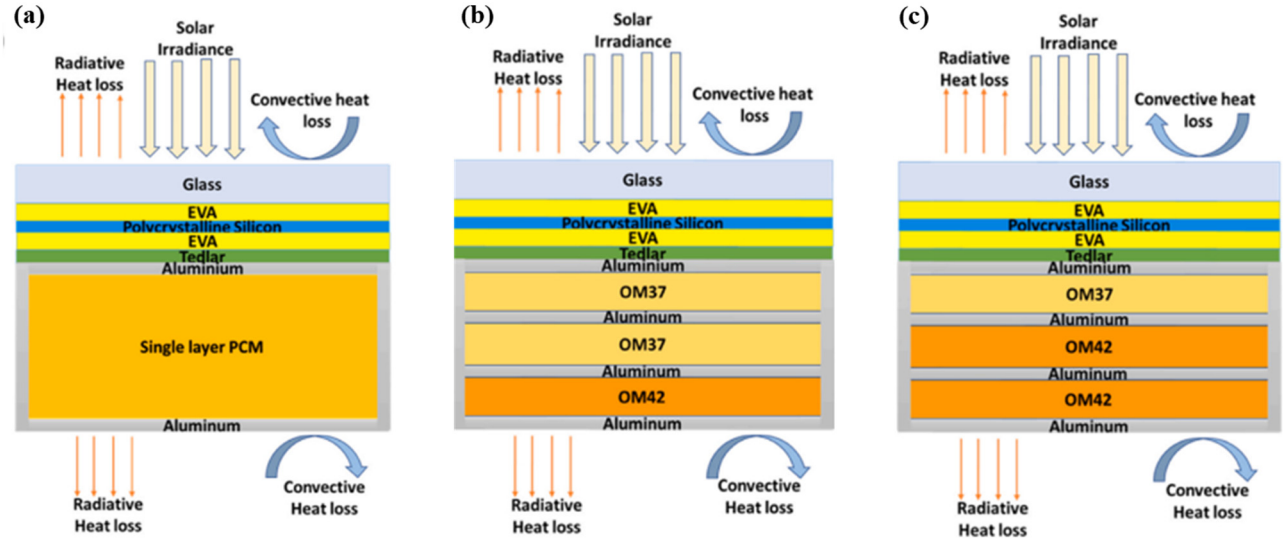


Figure 24. Diagrams of PV panels with multi-layer PCM cooling studied by Ranawa and Nalwa: (**a**) OM42, (**b**) OM37/OM37/OM42, and (**c**) OM37/OM42/OM42 [77].

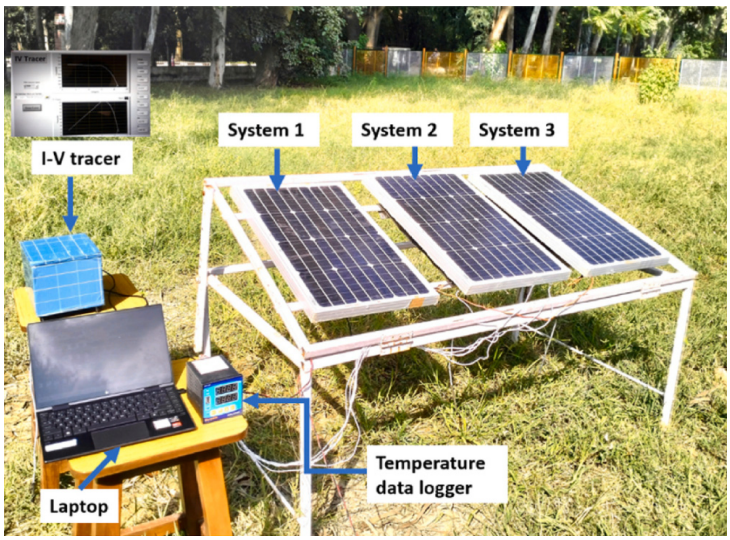


Figure 25. Experimental device for PV panels with multi-layer PCM cooling studied by Ranawa and Nalwa [77].

The traditional PCM cooling method for PV systems usually involves fixing a solid container filled with a PCM on the PV panel's back to regulate the panel temperature. Nizetic et al. [78] studied a PCM-based cooling approach for PV panels. They proposed to use several smaller PCM containers instead of a large PCM container and conducted some experiments. The layout of the small PCM containers is presented in Figure 26. The results showed that compared with uncooled PV panels, the power outputs of the PV panels with

a single large PCM container and with several small PCM containers increased by 2.5% and 10.7%, respectively. A similar study was also reported by Miaari and Ali [79].

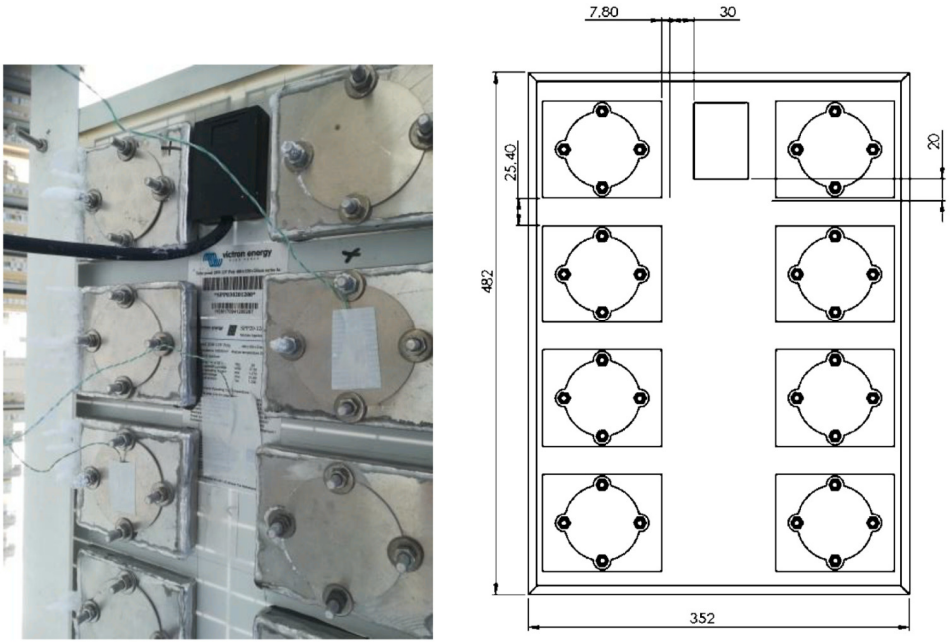


Figure 26. Photo and diagram of the layout of small PCM containers studied by Nizetic et al. [78].

Wongwuttanasatian et al. [80] conducted some experiments on cooling PV panels by using various kinds of PCM containers. The PCM used in the experiments was carnauba. The three evaluated PCM containers, namely, groove-type, tube-type, and fin-type containers, are presented in Figure 27. The comparison results showed that the fin-type container could induce the best cooling effect for PV panels. Compared with uncooled PV panels, the use of fin-type PCM containers could reduce the panel temperature from 57.9 °C to 51.8 °C and increase photo-electric efficiency from 9.33% to 9.82%.

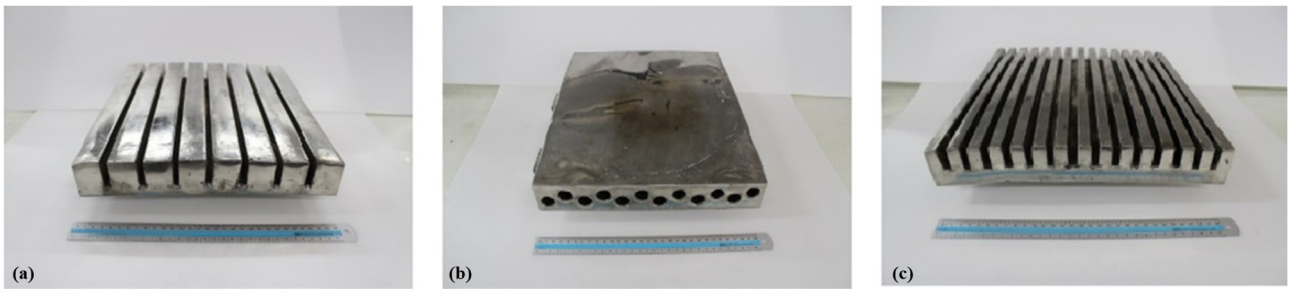


Figure 27. PCM containers: (**a**) groove, (**b**) tube, and (**c**) fin types [80].

Another method to enhance the cooling effect of PV panels with PCM cooling is adding fins at suitable positions in the PCM containers [81]. For instance, Qasem et al. [82] evaluated the cooling effect of paraffin wax on a CPV system (see Figure 28) through the simulation method. The PCM container was a cube box with T-shaped fins. The results indicated that compared with the conditions of installing fins in the middle and at the bottom, the PCM melting process of the CPV system with fins installed in the upper-half part could be enhanced by about 10.0%.

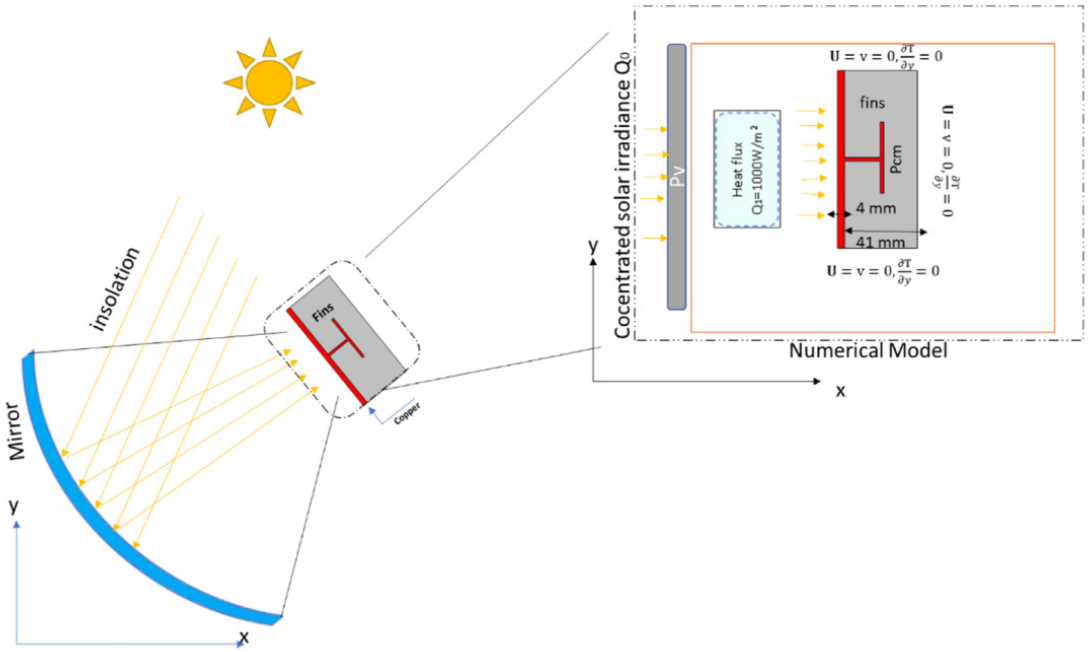


Figure 28. Diagram of a CPV system using PCM studied by Qasem et al. [82].

5.2. Composite PCM-Based Cooling

The cooling effects of PCMs can be limited by their low thermal conductivity [83]. An effective method has been proposed to solve this problem: adding particles with high thermal conductivity (e.g., nanoparticles) in PCMs [84] or coupling PCMs with other working fluids through some methods. Some studies showed that by adding various kinds of nanoparticles to PCMs, the thermal performance of PCMs can be improved and some thermophysical properties (e.g., latent heat, density and specific heat capacity) of PCMs can also be changed [85].

For instance, Aqib et al. [86] presented an experimental study on the physical properties of paraffin wax mixed with Al_2O_3 nanoparticles and with multi-wall carbon nano-tubes. The results demonstrated that doping nanoparticles could improve the thermal performance of paraffin wax. Prabhu et al. [87] experimentally prepared paraffin wax mixed with TiO_2/Ag composite nanoparticles for PV panel cooling. The field emission scanning electron microscope (FESEM) photo and X-ray diffraction test results of TiO_2/Ag composite nanoparticles prepared in their study are shown in Figure 29.

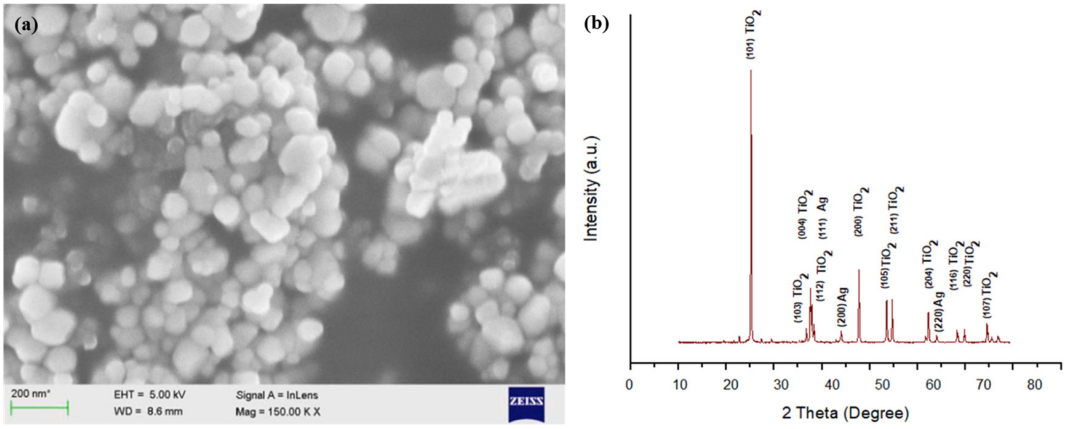


Figure 29. FESEM photo (**a**) and X-ray diffraction test results of TiO_2/Ag composite nanoparticles (**b**) in the study by Prabhu et al. [87].

Hassan et al. [88] compared four PV panels under different cooling conditions through the experimental method: PV panels with no cooling, with only a PCM (RT35HC), with water/RT35HC coupled cooling, and with nanofluid/RT35HC coupled cooling. The nanofluid was composed of graphene nanoparticles and water. Figure 30 presents the diagrams of pure PCM and nanofluid/PCM coupled cooling methods evaluated in the study. The results showed that compared with uncooled PV panels, the highest temperature reductions in the PV panels with nanofluid/ RT35HC, water/ RT35HC, and pure RT35HC were 23.9 °C, 16.1 °C, and 11.9 °C, respectively.

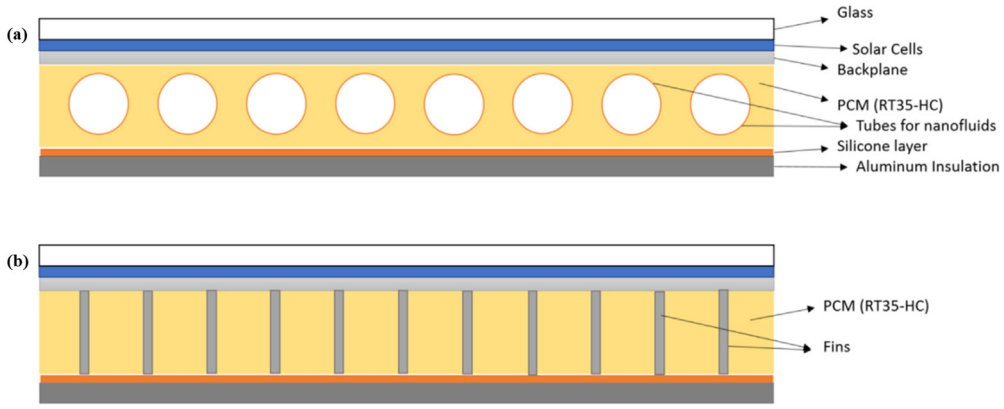


Figure 30. Diagrams of nanofluid/PCM coupled (**a**) and pure PCM (**b**) cooling methods for PV panels [88].

5.3. Brief Summary of PCM-Based Cooling Methods

The PCM-based cooling method is rather promising due to the advantages of PCMs [89]. For future research contents on PCM-based cooling, some suggestions are as follows:

(a) New low-cost dopant materials (e.g., multi-hole metal [90]) should be studied and searched, and the economic analysis of PV systems with new PCM cooling methods is also necessary.

(b) Local climate conditions can affect the utilization of PCMs in cooling PV panels [91]. The use of unsuitable PCMs may even damage PV panels. Hence, more experimental studies on larger-scale PV systems with PCM cooling should be conducted to investigate the usability of PCMs and overall performance of PV systems [92].

6. Spectrum-Based Cooling

6.1. Spectral Beam Splitting-Based Cooling

An important characteristic of PV cells is that they have spectral response curves [93]. This means that only a certain part of the full solar spectrum can be utilized by PV panels to generate electricity, and the rest absorbed by PV panels can only become waste heat and makes the panel temperature increase, thereby reducing photo-electric efficiency [94]. Though traditional PV/T systems can solve this problem to a certain extent, the temperature of the outlet working fluid is limited by the PV panel temperature [95]. Solar PV/T systems using beam splitting technology can not only reduce the PV panel temperature but also eliminate the limitation of PV panel temperature on working fluid temperature.

Spectral beam splitters mainly include two types, namely, solid and liquid beam splitters. Solid beam splitters are normally multi-layer optical films [96], and liquid beam splitters are spectrally selective liquids (e.g., water, nanofluid) [97]. Many studies on beam splitting PV/T systems have been conducted, and this subsection presents some typical studies.

The operating principle of solid beam splitter-based PV/T systems is shown in Figure 31. Regarding studies on beam splitting PV/T systems with solid beam splitters, Liang et al. [98] conducted experiments on a solar beam splitting PV/T system using a SiO_2/TiO_2 film beam splitter. They found that the use of SiO_2/TiO_2 film reduced the PV panel temperature by 3.0 °C.

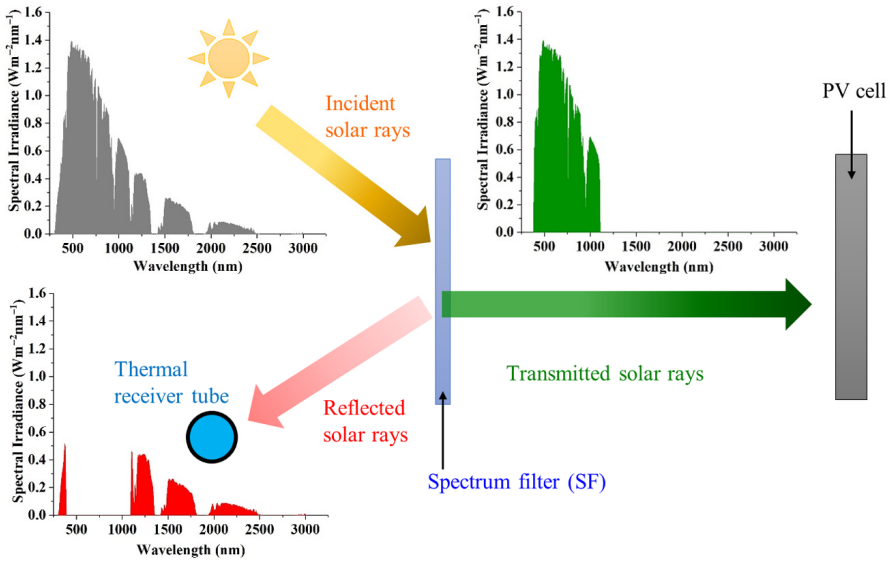


Figure 31. Operating principle of solid beam splitter-based PV/T systems.

Kandilli [99] carried out experimental and thermodynamic analyses on a solar beam splitting PV/T system using dish concentrators and hot mirror beam splitters. The experimental setup is presented in Figure 32. They found that the energy efficiency of that solar PV/T system was 7.3%, and the power generation cost was 6.37 USD/W.

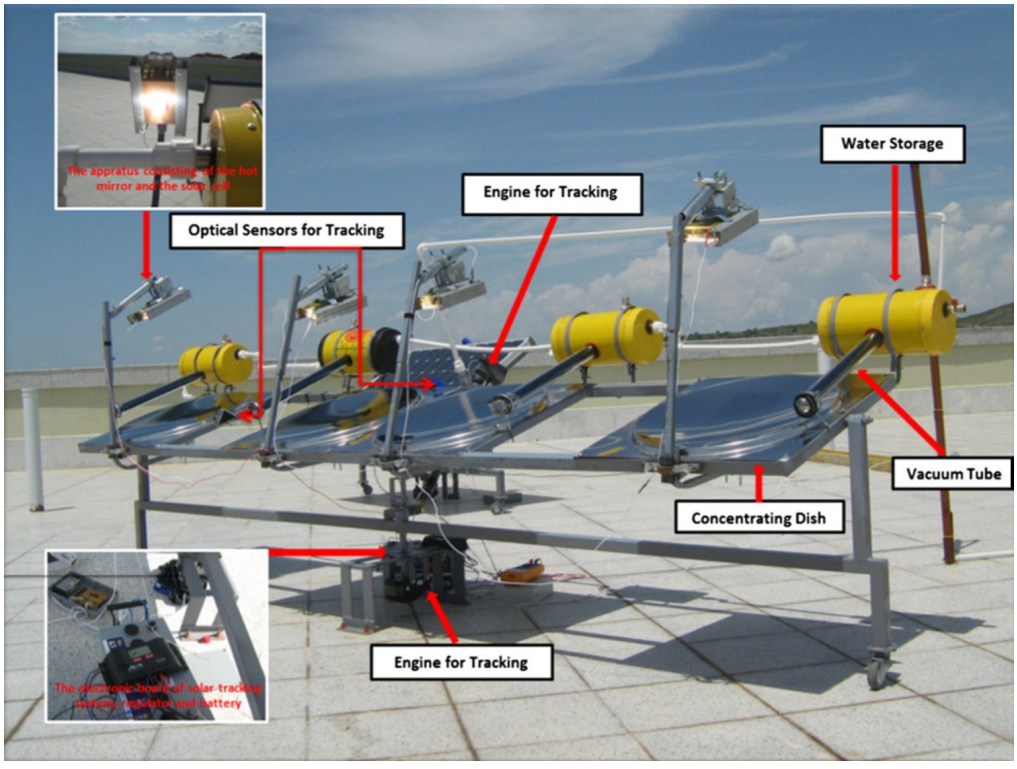


Figure 32. Experimental setup of a solar beam splitting PV/T system using dish concentrators and hot mirror beam splitters [99].

Wang et al. [100] proposed a solar beam splitting PV/T system using the compact linear Fresnel reflector (CLFR) and a $Na_3AlF_6/Nb_2O_3/Ge$ film beam splitter (see Figure 33). They evaluated the performance of that beam splitting PV/T system and found that the overall system efficiency was 26.7%, which was higher than that of the pure CPV system (23.8%).

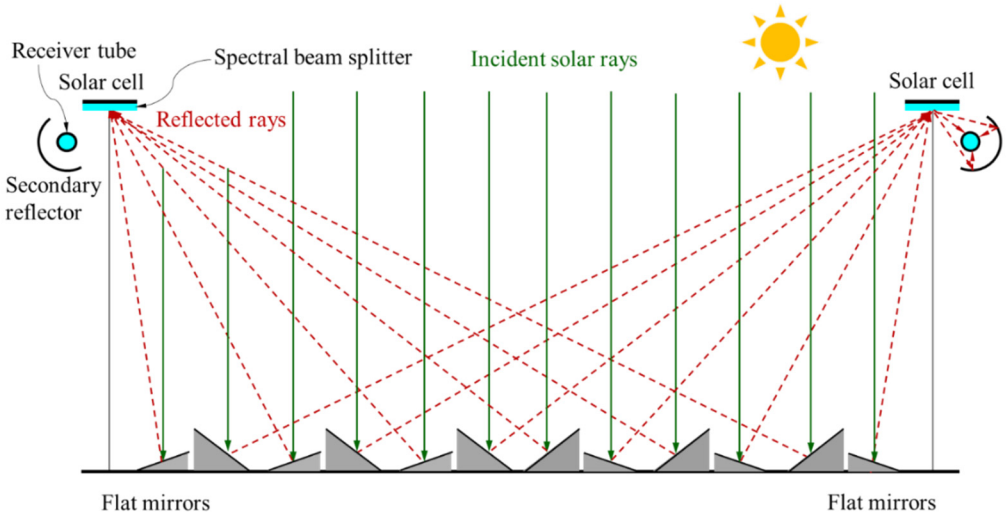


Figure 33. Diagram of a solar beam splitting PV/T system using CLFR [100].

Liquid beam splitters mainly use optical absorption to achieve beam splitting. Figure 34 presents the operating principle of liquid beam splitters [101]. Some common liquids, such as water and ethylene glycol, have certain optical absorption capability, but their optical absorption ranges are not easy to regulate and thus cannot match the spectral response ranges of PV cells well. When some kinds of nanoparticles are added in them, their spectral properties may be improved. Hence, many studies on solar beam splitting PV/T systems using nanofluids have been conducted [102].

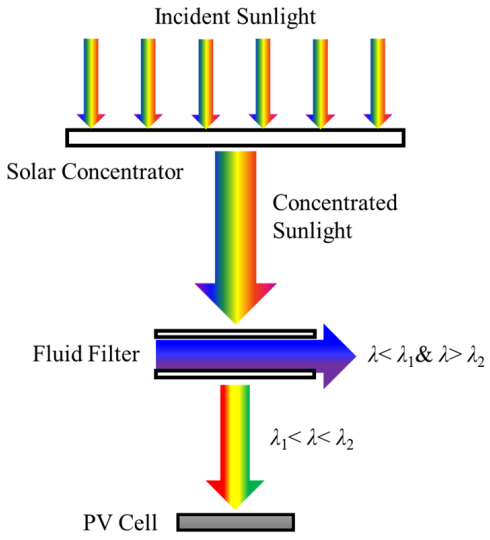


Figure 34. Operating principle of liquid beam splitters [101].

Abdelrazik et al. [103] studied a solar beam splitting PV/T system with a Ag/water nanofluid beam splitter. The influences of the runner height and flow rate of the nanofluid on the operation performance of the solar PV/T system were evaluated. The results revealed that the solar beam splitting PV/T system had better overall performance compared

with the pure PV system. When the environmental temperature was 25.0 °C, the electric and thermal efficiency rates of that solar beam splitting PV/T system were 10.7% and 57.7%.

Zhang et al. [104] studied an Ag/water nanofluid solar beam splitting PV/T system through the experimental method. The experimental device of the solar beam splitting PV/T system is presented in Figure 35. Indoor experiments were conducted by using a solar simulator. The results revealed that using the Ag/water nanofluid beam splitter is an effective method to regulate PV panel temperature. When the incident solar intensity was 900.0 W/m^2, the nanofluid thickness changed from 10.0 mm to 30.0 mm and the PV panel temperature decreased from 34.5 °C to 33.0 °C.

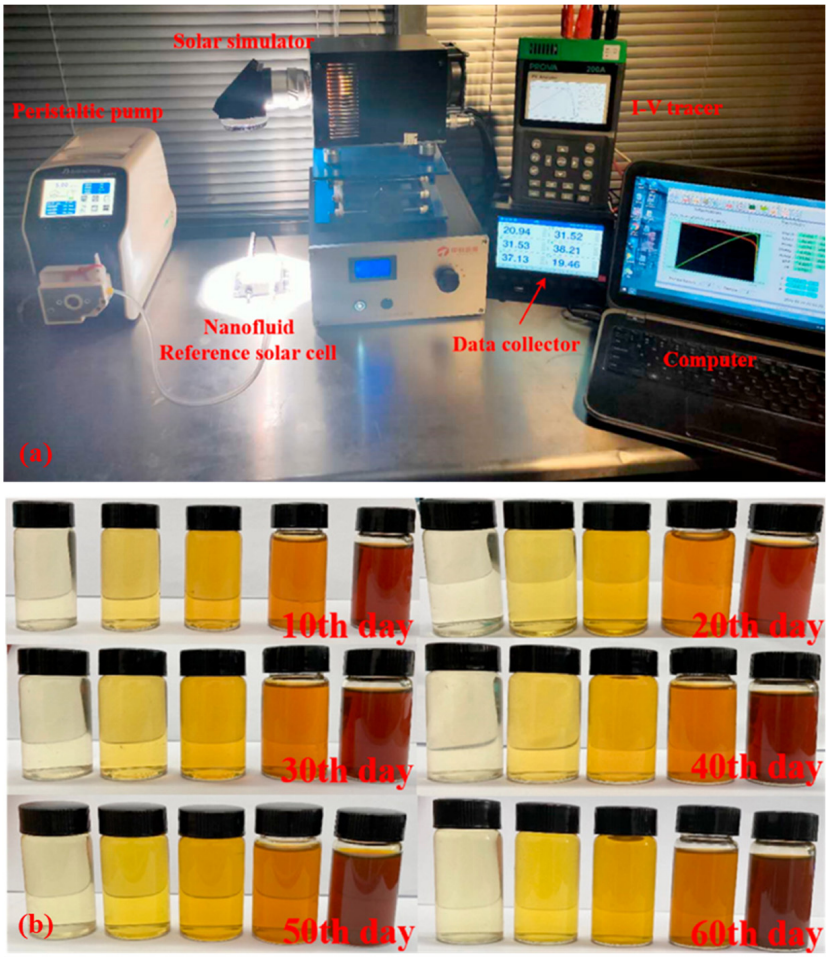


Figure 35. Experimental device of a solar PV/T system using Ag/water nanofluid beam splitter (**a**) and photos of Ag/water nanofluids with different Ag concentrations at different times (**b**) [104].

6.2. Radiational Cooling

The Stefan–Boltzmann law states that all objects have the property of thermal radiation, and their thermal radiation power is in positive proportion to the fourth power of their surface temperatures. Water vapor and CO_2 contained in the atmosphere can hinder radiation heat exchange between the objects on the earth surface and the outer space, but the thermal radiation of the 8.0~14.0 μm waveband (i.e., atmospheric window) can have very high transmissivity in the atmosphere, allowing the objects on the earth surface to release thermal energy to outer space [105]. When the thermal radiation release of an object is higher than the incident thermal energy flux, radiational cooling arises.

Current studies on radiational cooling methods mainly focus on two aspects, namely, incident solar reflection and mid-infrared radiation emission. The reflection of incident

sunlight can be achieved through metal-layer reflection [106], micro–nanoparticle scattering [107], porous scattering [108], etc. And according to whether the infrared emission spectrum is within 8.0~14.0 μm, radiation emitters are divided into selective emitters [109] and broad-spectrum emitters [106]. It has been reported that selective emitters have greater advantages in the field of low-temperature radiational cooling, while broad-spectrum emitters have greater advantages under conditions near ambient temperature.

Many studies on nocturnal radiational cooling which reveal the feasibility of nocturnal radiational cooling methods to a certain extent have been conducted [110]. However, the incidence of solar radiation makes it more challenging to achieve radiational cooling in daytime. With the development of materials science, some new materials, which have both relatively high reflectance in some solar spectral ranges and high emissivity in 8.0~14.0 μm, have been developed, thereby breaking through the limitations of daytime solar radiation [111].

When radiational cooling methods are utilized to cool PV panels, the radiational cooling materials should not hinder the absorption of solar energy in 0.38~1.1 μm and should have high emissivity in the spectrum range of 8.0~14.0 μm. Radiational cooling methods for PV panels have attracted broad attention [112]. Figure 36 presents the diagram of the radiational cooling of a PV panel.

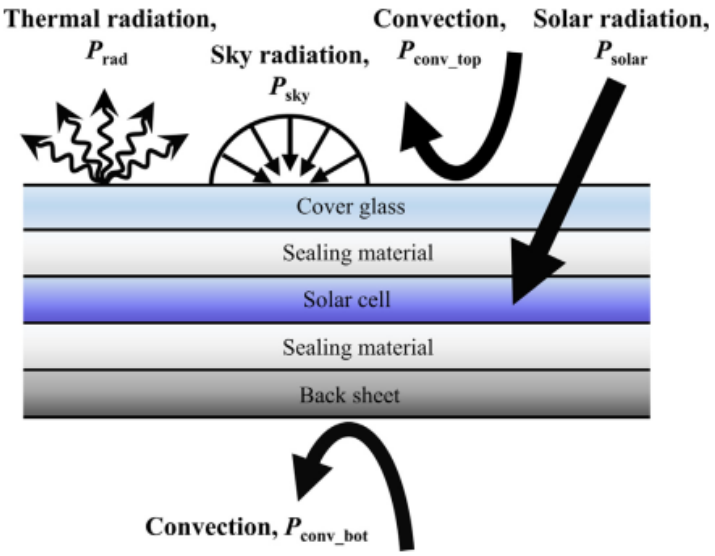


Figure 36. Diagram of the radiational cooling of a PV panel [105].

Zhao et al. [113] evaluated the effect of radiational cooling on the performance of commercial PV modules and discussed the application potential of a radiational cooling method. They achieved radiational cooling by plating polydimethylsiloxane (PDMS) on PV panels. Figure 37 presents a photo of the coated PV panel. The results showed that by using PDMS, the panel temperature decreased by about 1.8 °C.

Long et al. [114] studied a radiational cooling method of using SiO_2 micro-grid coating material for PV panels. By using heat transfer simulations, the radiational cooling performance of the PV panels was evaluated. The results showed that compared with uncooled PV panels, the panel temperature reduction brought by the micro-grid coating material could be up to 20.0 °C.

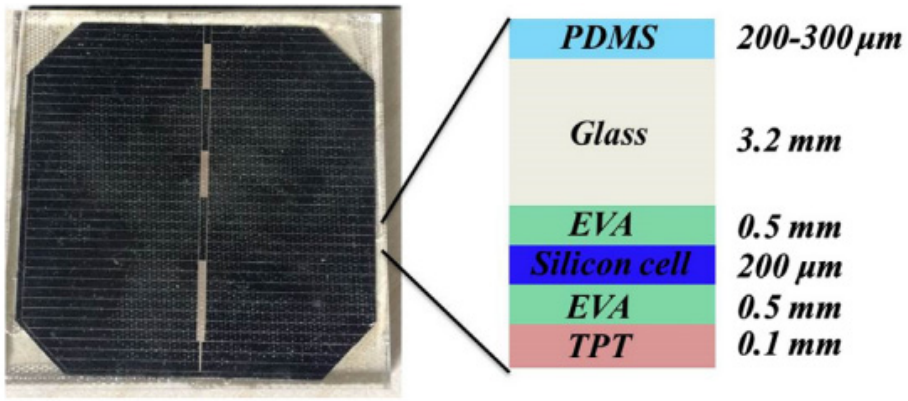


Figure 37. PDMS-coated PV panel [113].

The radiational cooling materials used for PV panels reported in existing research have been plated on the PV panel upper surface. Many of them enhance the emission of mid-infrared radiation while hindering the absorption of sunlight by PV panels to a certain extent. Regarding this issue, Li et al. [115] proposed an indirect radiational cooling PV system composed of PV panels, a radiational cooling block, a cold storage block, and pipes. Water served as the working fluid and the carrier of cold. Figure 38 presents a photo of the indirect radiational cooling PV system. Experiments were conducted, and the results showed that on a typical summer day, compared with uncooled PV panels, the indirect radiational cooling method reduced the panel temperature by 17.8 °C.

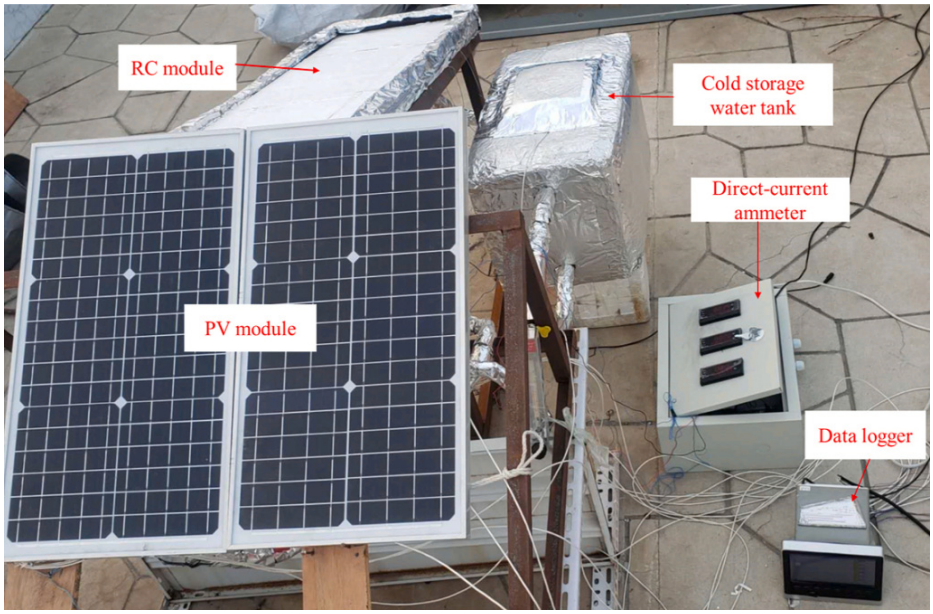


Figure 38. Photo of an indirect radiational cooling PV system [115].

Compared with glass-sealed PV panels, even when a coating with ideal radiation emissivity (1.0) is plated on the glass upper surface, the panel temperature cannot be reduced more significantly [116]. Regarding this problem, Ahmed et al. [117] designed a PV system with better radiational cooling effect which comprised a PV power block, a radiational cooling block, and a heat pipe (see Figure 39). The PV and radiational cooling blocks were arranged on the evaporation and condensation sections of the heat pipe, respectively. Through the simulation method, the operation performance of the PV system was evaluated. The results indicated that compared with traditional glass-sealed PV panels, the panel temperature decreased by 12.9 °C.

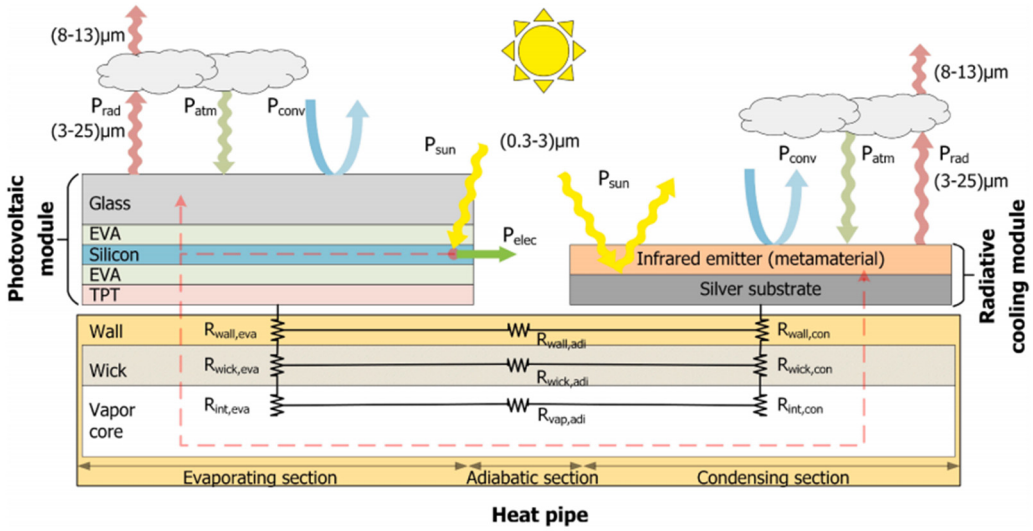


Figure 39. Diagram of a PV system with radiational cooling and a heat pipe [117].

6.3. Brief Summary of Spectrum-Based Cooling Methods

Theoretically, beam splitting PV/T technologies can reduce the PV panel temperature effectively while improving the overall solar energy efficiency. But there are still some unresolved problems. For instance, the spectral properties of current beam splitters cannot match that of ideal beam splitters for different kinds of PV cells very well, and the development of new beam splitters is still necessary. Additionally, some beam splitting PV/T systems are still structurally complex and not suitable for large-scale production. Thus, future layout design and experimental evaluations of new beam splitting PV/T systems will also be meaningful.

Currently, radiational cooling technology is still in the experimental stage, and relevant further research contents may include the following [118]:

(a) The design, research, and manufacturing of economically feasible large-scale radiational coolers for PV systems are still in need.

(b) Further evaluations should be conducted on the applicability of radiational cooling methods, especially daytime radiational cooling methods, to seasons and geographical locations.

7. Evaporative Cooling

Evaporative cooling is a method for controlling overheating [119]. When a small portion of the coolant undergoes a phase change and takes away excess heat from the surface, evaporative cooling occurs [120]. Evaporative cooling methods have the characteristics of high cost-effectiveness, easy implementation, effectiveness under dry climate conditions, and ability to operate in a stable manner [121]. Relative humidity, temperature, air flow, and exposed surface area can all affect evaporation [122]. In recent years, many experimental and simulation studies on evaporative cooling methods for PV panels have been conducted. Existing evaporative cooling systems mainly use materials such as synthetic clay [123], cotton wicks [124], cloth and rubber pipes [121], zeolite [125], porous layers [126], fabric and sackcloth [127], fiber pads or fiber [128], and hydrophilic pads [129]. This section provides some typical reported studies.

Lucas et al. [130] evaluated the feasibility of utilizing evaporative cooling for PV panels. A system which consisted of PV panels, water evaporation circulation, and a water condensation cooler (see Figure 40) was proposed. Through experimental tests, the thermal and electrical performance of the system was analyzed. The results indicated that at noon in summer, the maximum increase in the electric efficiency of the PV panel was 7.6%.

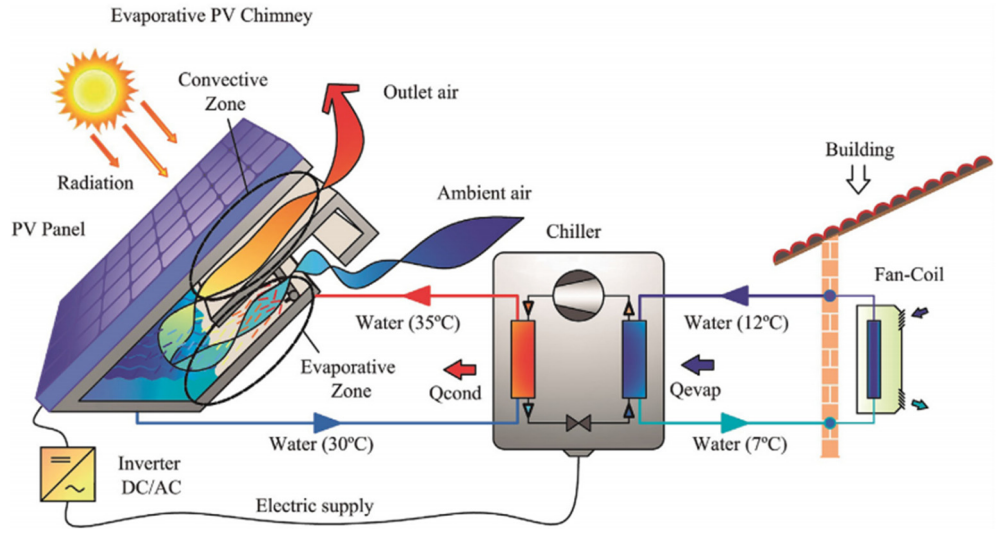


Figure 40. A PV system using evaporative cooling and a water condensation cooler [130].

Srithar et al. [131] investigated a method of cooling PV panels with evaporation and a solar distiller through the experimental method. A sackcloth was attached on the PV panel's back and its two ends were immersed in water. The results revealed that by using the proposed method, the panel temperature decreased by 8.0 °C and the output power and electric efficiency of the PV panel increased by 5.6% and 14.5%, respectively. And Dida et al. [132] conducted a similar experimental study.

Zizak et al. [128] evaluated the performance of an evaporative cooling method for PV panels under different climatic conditions through experiments. Figure 41 presents the experimental device. The results revealed that compared with uncooled PV panels, the cooling method reduced the panel temperature by 20.1 °C.

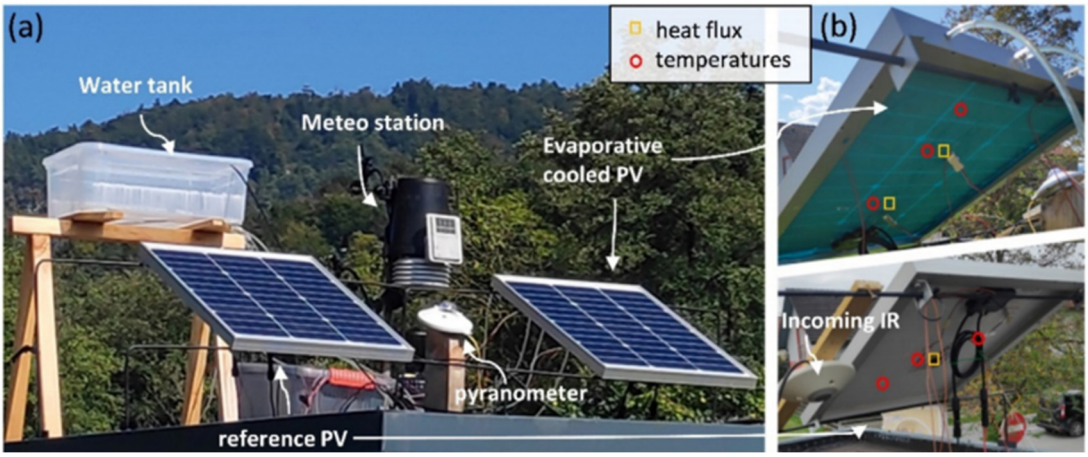


Figure 41. Experimental device for PV panels with and without evaporative cooling (**a**) and the back-surface views of the two PV panels (**b**) [128].

Abed et al. [133] conducted a simulation study on the evaporative cooling of PV panels by using COMSOL software. A wet porous wick was assumed to be attached on the back of PV panels, forming a simple water passive cooling device. The results showed that the panel temperature decreased by 9.6%.

Hydrogel composites have attracted much attention in atmospheric water collection due to their excellent moisture absorption capacity [134]. As the cooling material, hydrogel can dissipate the waste heat generated in PV panels through water evapora-

tion [135]. Many studies on hydrogel-based evaporative cooling methods for PV panels have been conducted.

Liu et al. [136] conducted an experimental evaluation of a PV panel with a cooling layer composed of polyacrylamide, carbon black, and LiCl. Carbon black and LiCl were added in the polyacrylamide hydrogel, and the composite was used as the moisture absorbent. The indoor experimental setup is shown in Figure 42. The results showed that when the radiation intensity was 1000.0 W/m^2, hydrogel cooling reduced the panel temperature by 9.9 °C and thus increased photo-electric efficiency by approximately 5.9%.

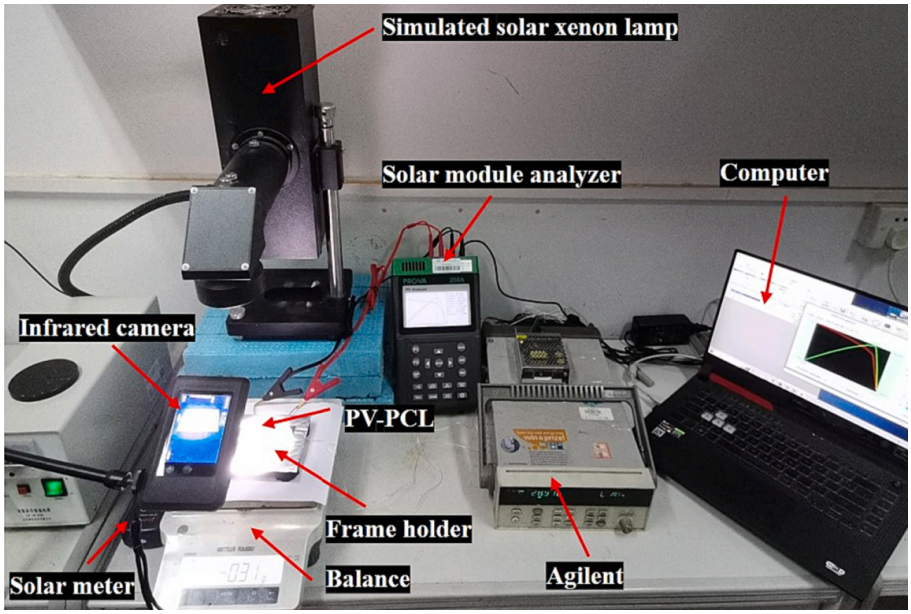


Figure 42. Indoor experimental setup of a PV panel using polyacrylamide supplemented with carbon black and LiCl [136].

Zou et al. [137] analyzed a PV panel cooling method based on the combined utilization of fins and hydrogel. The hydrogel was composed of polyacrylamide and LiCl. Preparations of polyacrylamide/LiCl material and cooling effect tests of different cooling methods on PV panels were conducted. Figure 43 presents the SEM photo of freeze-dried polyacrylamide hydrogel and some other photos of experimental samples. The test results revealed that when the environmental temperature and relative humidity were 25.0 °C and 70.0%, after sixteen hours of moisture absorption, the water absorption ability of the prepared hydrogel was 1.87 g/g. Figure 44 provides some infrared radiation test results of different cooling structure surfaces at different heating times.

According to the existing studies, some unsolved problems for evaporative cooling still exist. Further studies on evaporative cooling methods for PV systems may include the following:

(a) Design and experimental tests of evaporative cooling systems with water collection and re-circulation devices will be meaningful.

(b) More effective moisture-absorbing materials used in evaporative cooling systems should be studied and developed.

(c) The development of new alternative coolants with low evaporation temperature and high thermal capacity should be conducted.

(d) For future practical applications, it is also necessary to conduct design optimization studies to make evaporative cooling systems more suitable for integrated processing with PV panels.

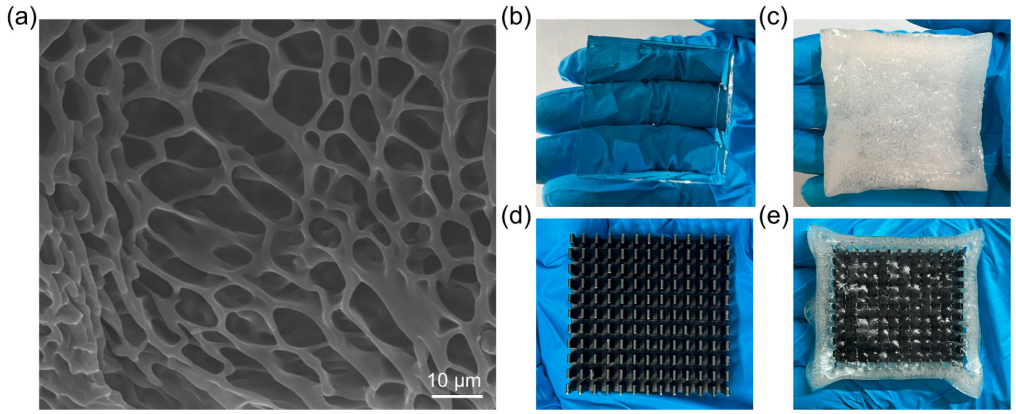


Figure 43. SEM photo of freeze-dried polyacrylamide (**a**), photo of swollen polyacrylamide hydrogel (**b**), photo of polyacrylamide/LiCl (**c**), photo of fin heat sink (**d**), and combination of fins and composite hydrogel (**e**) [137].

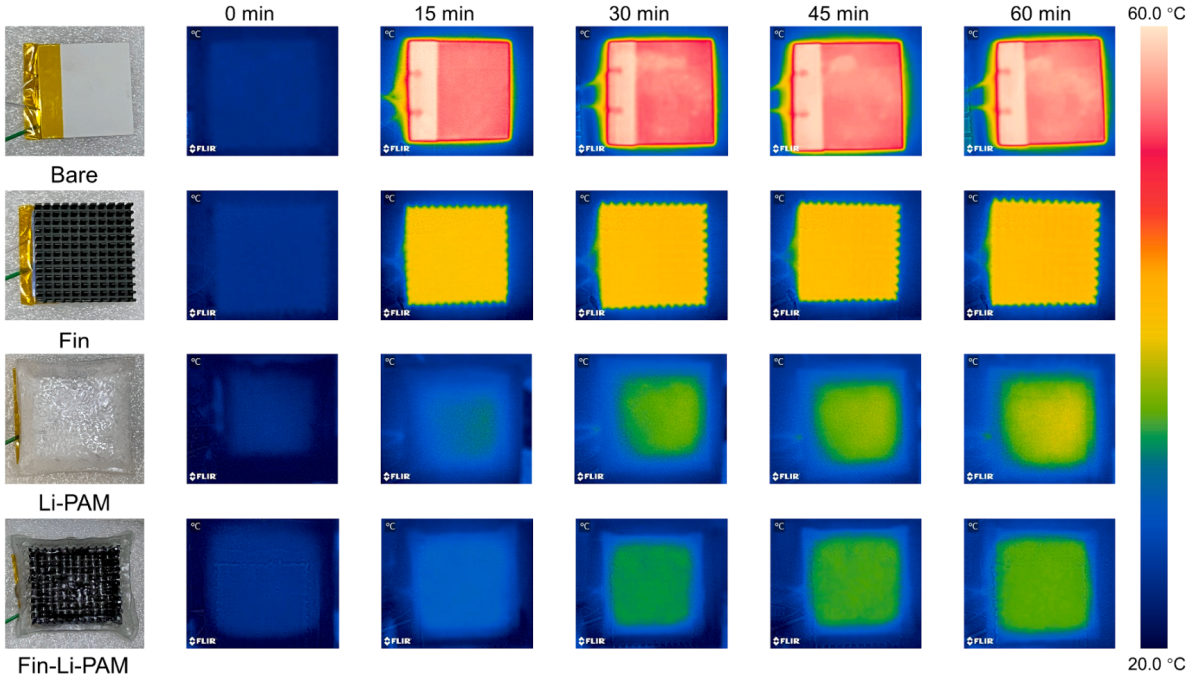


Figure 44. Infrared radiation test results of bare sheet, fin heat sink, hydrogel, and fin/hydrogel structure surfaces at different heating times [137].

8. Thermo-Electric Cooling

The thermo-electric cooling (TEC) phenomenon is based on the Peltier effect. In TEC devices, the thermo-electric elements are usually made of a special semiconductor material called thermo-electric material [138]. When current passes through thermo-electric materials, the hot end of the thermo-electric element absorbs heat, and the cold end releases heat. This makes the cold-end temperature lower than that of the hot end, creating a cooling effect. Thermo-electric modules can be utilized to collect waste heat generated by PV panels and convert it into electrical energy [139].

Some feasibility studies on integrating schemes of TEC devices and PV panels have been conducted. For instance, Benghanem et al. [140] conducted an evaluation on a TEC method for PV panels through the experimental method. Figure 45 presents the experimental setup. The results showed that the temperature of the uncooled panel was

83.0 °C and that of the panel with TEC was reduced to 65.0 °C, showing significant cooling effect.

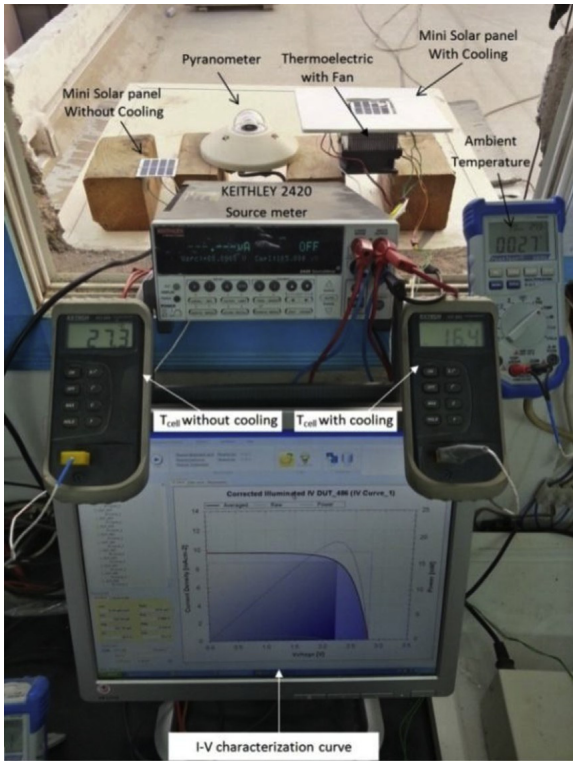


Figure 45. Experimental setup of a PV panel with a thermo-electric module established by Benghanem et al. [140].

Hamzehzarghani and Eslami [141] carried out a simulation evaluation on the feasibility of TEC used in a PV system under different conditions. They found that when the optimum current was provided to the thermo-electric module, compared with uncooled PV panels, a marginal power increase could be obtained.

Salari et al. [142] proposed a PV/T system with TEC (see Figure 46) and conducted thermal and electrical performance evaluations on the system through simulations. The results revealed that when the solar intensity was 1000.0 W/m^2, the electric efficiency of the PV/T system with TEC was 10.4% higher than that of the pure PV/T system.

Kane et al. [143] conducted the optimization of a TEC method for PV panels through simulations. The results indicated that when temperature-based maximum power point tracing control was used, with solar intensity variation within 800.0~1000.0 W/m^2 and environmental temperature variation within 25.0~45.0 °C, the PV panel temperature reduction was 6.0~26.0 °C.

Currently, the cooling efficiency of TEC methods is relatively low, making TEC methods only suitable for reducing the temperature of small-scale PV panels. Further studies may include the following:

(a) Some studies on the combination of TEC and other passive cooling methods (heat pipes, beam splitting, etc.) are recommended to be conducted in the future.

(b) Optimization and economic evaluation of TEC devices used in PV systems are important and necessary.

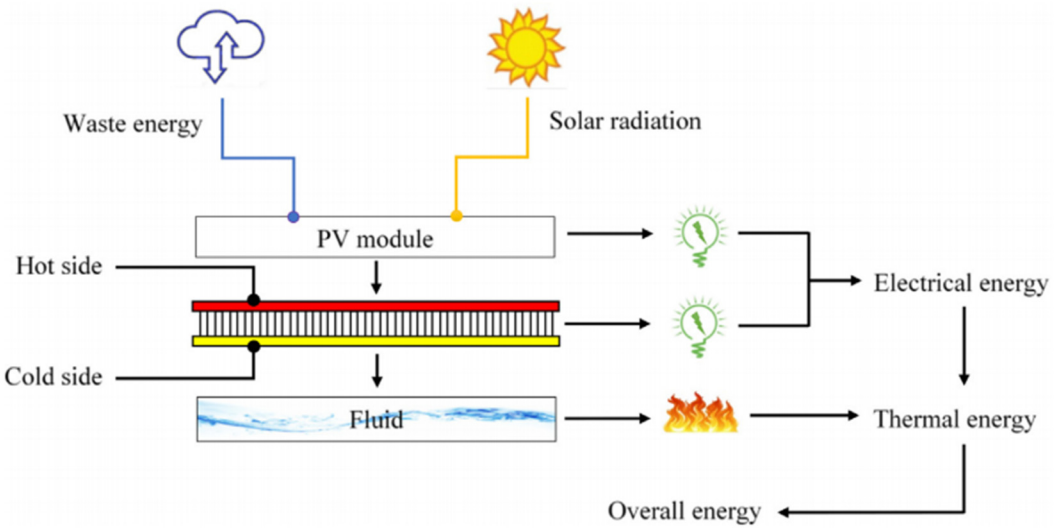


Figure 46. Diagram of a PV/T system with TEC [142].

9. Composite Cooling

Under some conditions, the combined utilization of two or more cooling methods can generate better cooling effects, improving the photo-electric efficiency and service life of PV systems [144]. This section presents some typical studies on composite cooling methods for PV panels.

Sudhakar et al. [145] studied a composite cooling method combining PCM and water natural circulation for PV panel cooling (see Figure 47). Experiments on the composite cooling method were conducted. A PCM (OM35) was attached on the PV panel's back, and water flowed in the channel attached on the back of the PCM. The results showed that compared with uncooled panels, the average power generation, electric efficiency, average panel temperature decrease, total exergic output, and exergic efficiency of the panel with composite cooling increased by 11.9%, 12.4%, 5.4 °C, 26.1%, and 8.1%, respectively.

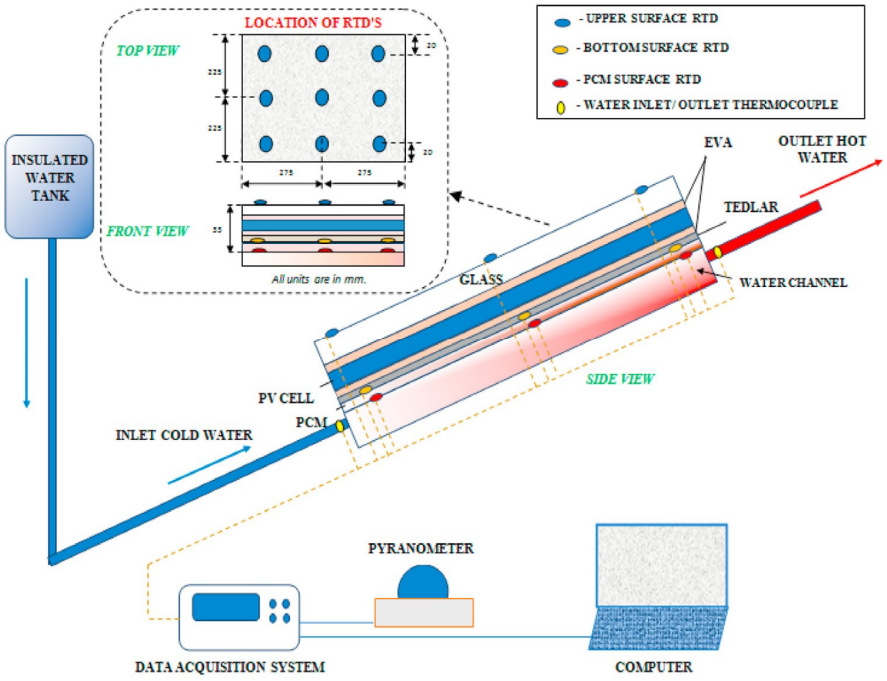


Figure 47. Diagram of experimental device of a PV panel with PCM and water natural circulation combined cooling [145].

Cui et al. [146] proposed a composite cooling method for concentrated solar PV panels which combined a PCM, a thermo-electric device, and a heat sink (see Figure 48). Different coolant materials could be used in the heat sink. Experiments were conducted to evaluate the feasibility of the composite cooling method. The results revealed that the composite cooling method could reduce the panel temperature and increase the output power of the PV system. When the solar concentration ratio was 1156, by using air in the heat sink, the maximum output power of the PV system using the proposed composite cooling method was 8.2 W, which was 2.0 W higher than that of an uncooled PV system.

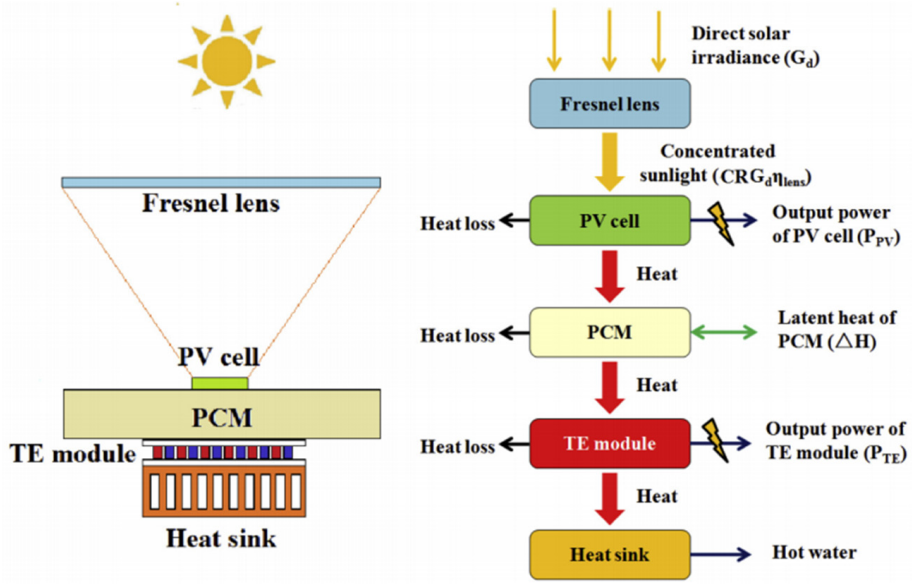


Figure 48. Diagram of a CPV system using combined cooling with a PCM, a thermo-electric device, and a heat sink [146].

Yusuf [147] proposed a composite cooling method for PV panels which can make a PV system export electric power continuously over a whole day. The composite cooling method is a combination of a PCM, a thermo-electric device, and radiational cooling (see Figure 49). Simulations on the PV system were conducted. The results showed that the PV system could operate continuously in the long term with none or very little maintenance. For the summer of Helsinki, the daytime and nighttime highest output power values of the PV panel with the composite cooling method were 120.0 mW and 70.0 μW, respectively.

In general, a proper combination of multiple cooling methods could enhance the cooling effect on PV panels. However, some further studies should be considered, as follows:

(a) Numerical and experimental studies are both necessary for evaluating the technical feasibility of newly designed composite cooling methods for PV panels.

(b) Economic analysis of newly designed composite cooling methods for PV panels should also be carried out to investigate their economic feasibility or to provide some critical economic parameter requirements for them.

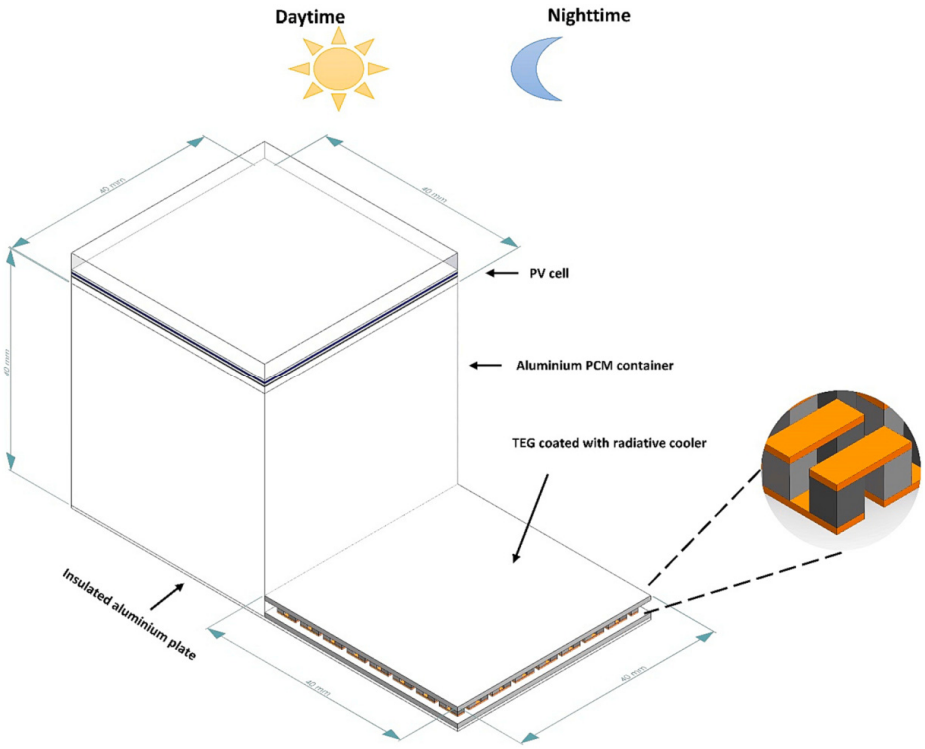


Figure 49. Diagram of a PV panel with the combination of a PCM, a thermo-electric device, and radiational cooling [147].

10. Conclusions

This paper conducts a comprehensive review of research works on eight different types of cooling methods for PV panels, i.e., air, liquid, heat pipe, PCM-based, spectrum-based, evaporative, thermo-electric, and composite cooling methods. Typical simulation and experimental studies on the eight types of cooling methods are introduced, and advantages, existing problems, and further research suggestions for these cooling methods are also discussed, aiming to provide a reference for future studies on and developments in PV panel cooling technologies.

Table 1 provides a brief summary of the eight types of cooling methods for PV panels, presenting cooling efficiency, cost implication, scalability, complexity, and future research directions for different types of cooling methods. In Table 1, issues related to the cooling efficiency, cost implication, scalability, and complexity of different PV panel cooling methods are included in the Advantages and Limitations columns. Figure 50 presents a brief roadmap for future research directions of different PV panel cooling methods. Different types of cooling methods have different advantages, existing limitations and future research focuses. Here, some general suggestions for future research directions for PV panel cooling methods are provided as follows:

(a) Research and development of new materials used in PV panel cooling methods should be continuously conducted, for instance, new HTF materials used in active cooling, new composite PCMs, new solid and liquid beam splitters used in beam splitting-based cooling, and new moisture-absorbing materials used in evaporative cooling.

(b) Studies on new combinations of multiple PV panel cooling methods are meaningful. Operation simulation, experimental evaluation, and optimization analysis on newly developed composite cooling methods considering climatic and geographical conditions are all necessary.

(c) The economy of a system is important, as it directly impacts the application feasibility of the system. The main challenge in commercializing these PV panel cooling

methods is whether the benefits from the increased power generation can surpass the increase in investment and O&M costs. For all studied and developed PV panel cooling methods, economic analysis should be conducted to evaluate their economic feasibilities.

Table 1. Brief summary of different types of cooling methods for PV panels.

Cooling Method Types	Advantages	Limitations	Future Research Directions
Air cooling	(a) Initial investment cost is lower. (b) Cooling structures are relatively less complex. (c) O&M works are simple. (d) Risks associated with freezing or leakage can be reduced.	(a) Natural air cooling relies on air flow; thus, its heat dissipation effect is greatly influenced by environmental factors. (b) Forced circulation air cooling relies on external power, reducing the net output power of PV panels.	(a) More studies on enhancing passive natural convection air cooling methods should be conducted, mainly focusing on the optimization of fin parameters and arrangement, design of new air flow channels, and new structural renovations of PV panels. (b) For forced circulation air cooling methods, some economic evaluations on the relationship between the power increase in a PV system and the external power consumption by the cooling system should be conducted. (c) Studies on air cooling methods for PV panels in high-temperature environments (e.g., deserts or hot summers) are meaningful.
Liquid cooling	(a) Compared with air cooling, liquid cooling can normally further enhance cooling efficiency in PV panels. (b) Liquid cooling is suitable for PV cells of different types and sizes and has strong versatility and flexibility. (c) O&M costs of floating PV systems are relatively low.	(a) Liquid cooling requires electrical insulation between the PV panel and the coolant, increasing the complexity of the system. (b) A risk of liquid leakage exists in some liquid cooling systems. (c) Active liquid cooling methods need external power, reducing the net output power of PV panels. Relatively higher maintenance cost is also needed.	(a) For forced liquid cooling methods, further studies may include heat transfer enhancement by modifying liquid flow channels, optimization of spray nozzle structure and arrangement, pulse jet or spray cooling, and economic feasibility analysis of cooling methods. (b) For liquid immersion cooling, further studies on PV panel temperature uniformity analysis and evaluation of immersion depth's effect on the operation performance of PV panels will be meaningful. (c) For floating PV systems, future studies include optimization of PV panel arrangement, cooling performance evaluations of thin-film PV panels used in floating PV systems, and economic feasibility analysis. In addition, some issues unrelated to PV panel cooling should also be considered, such as the evaluation of the effect of water corrosion on the structure and operation of PV panels and the analysis of the effect of floating PV systems on ecological footprint, water quality, and other environmental factors.

Table 1. *Cont.*

Cooling Method Types	Advantages	Limitations	Future Research Directions
Heat pipe cooling	(a) Heat pipe cooling needs no large temperature gradient between the heat source and heat sink. It can have high temperature uniformity, high thermal conductivity, and variable heat flux without additional energy consumption. (b) Low initial cost, high reliability, and long service life. (c) Heat pipes have simple and diverse structures and can be freely designed according to the heat dissipation requirements and structural characteristics of different PV systems.	(a) Heat pipes require the regular maintenance and cleaning, increasing maintenance and time costs. (b) Long-term contact between working fluid and pipeline wall may lead to corrosion, affect the efficiency of gas–liquid circulation, and even cause small cracks in the pipe, resulting in the evaporation of the working fluid.	(a) The complex latent heat transfer and convective heat transfer mechanisms inside some kinds of heat pipes (e.g., pulsating heat pipes) still need further exploration. (b) The evaluation of the effect of heat pipe cooling on PV panel temperature uniformity is necessary in the future. (c) The efficiency research, economic cost–benefit analysis, and environmental benefit evaluation of long-term operation of heat pipe-cooled PV systems under actual conditions should be conducted.
PCM-based cooling	(a) PCM cooling utilizes phase change to absorb heat and can dissipate heat efficiently. (b) PCM cooling is passive and can save energy. (c) Environmentally friendly.	(a) Using unsuitable PCMs can have different cooling results and even damage PV panels. (b) PCMs may experience performance degradation during long-term use, leading to a decrease in cooling efficiency and affecting the stability of the PV system.	(a) New low-cost dopant materials (e.g., multi-hole metal) should be studied and researched, and the economic analysis for PV systems with new PCM cooling methods is also necessary. (b) New PCMs should be explored and evaluated for PV panel cooling. (c) More experimental studies on larger-scale PV systems with PCM cooling should be conducted.
Spectrum-based cooling	(a) Spectral beam splitting-based cooling can effectively reduce PV panel temperature and overcome the temperature limitation of the panel on the temperature of the working fluid, leading to a higher thermal grade and higher overall energy efficiency. (b) The advantages of radiational cooling lie in its simple structure and environmental friendliness. Moreover, radiational cooling materials are lightweight and can effectively reduce PV panel temperature without consuming other energy.	(a) Currently, the structures of solar beam splitting PV/T systems are normally relatively complex, and the design of more ideal beam splitters is still difficult. (b) Radiational cooling methods are greatly influenced by the geographical and climatic conditions.	(a) The development of new, more ideal beam splitters is still necessary. (b) Layout designs and experimental evaluations of new solar beam splitting PV/T systems are meaningful future efforts. (c) The design, research, and manufacturing of economically feasible large-scale radiational coolers for PV systems are still needed. (d) Further evaluations should be conducted on the applicability of radiational cooling methods, especially daytime radiational cooling methods, to seasons and geographical locations.

Table 1. *Cont.*

Cooling Method Types	Advantages	Limitations	Future Research Directions
Evaporative cooling	(a) Evaporative cooling systems normally have simple structures and are easy to implement. (b) Lower investment cost. (c) Suitable for cooling PV panels in dry climates. (d) Environmentally friendly.	(a) Evaporative cooling devices need regular maintenance and cleaning. (b) Some evaporative cooling methods need a certain amount of water, which may create certain pressure on water resources in water-scarce areas. (c) Hydrogels have some disadvantages, including low thermal conductivity and structural instability. And they are greatly affected by environmental humidity.	(a) Design and experimental tests of evaporative cooling systems with water collection and re-circulation devices will be meaningful. (b) More effective moisture-absorbing materials used in evaporative cooling systems should be studied and developed. (c) The development of new alternative coolants with low evaporation temperature and high thermal capacity should be conducted. (d) It is necessary to conduct design optimization research to make evaporative cooling systems more suitable for integrated processing with PV panels.
TEC	(a) No need for coolant. (b) Low energy consumption. (c) Good stability.	(a) Low cooling efficiency.	(a) Some studies on the combination of TEC and other passive cooling methods (heat pipe, beam splitting, etc.) are recommended to be conducted in the future. (b) Optimization and economic evaluation of TEC devices used in PV systems are important and necessary.
Composite cooling	(a) Proper combination of two or more cooling methods may enhance cooling effect on PV panels. (b) Some composite cooling schemes can make PV systems operate continuously and stably for the long term.	(a) Composite cooling may need higher investment cost. (b) More O&M work and cost are needed. (c) Composite cooling may bring relatively complex system structures.	(a) Numerical and experimental studies are both necessary for evaluating the technical feasibility of newly designed composite cooling methods for PV panels. (b) Economic analysis for newly designed composite cooling methods for PV panels should also be conducted.

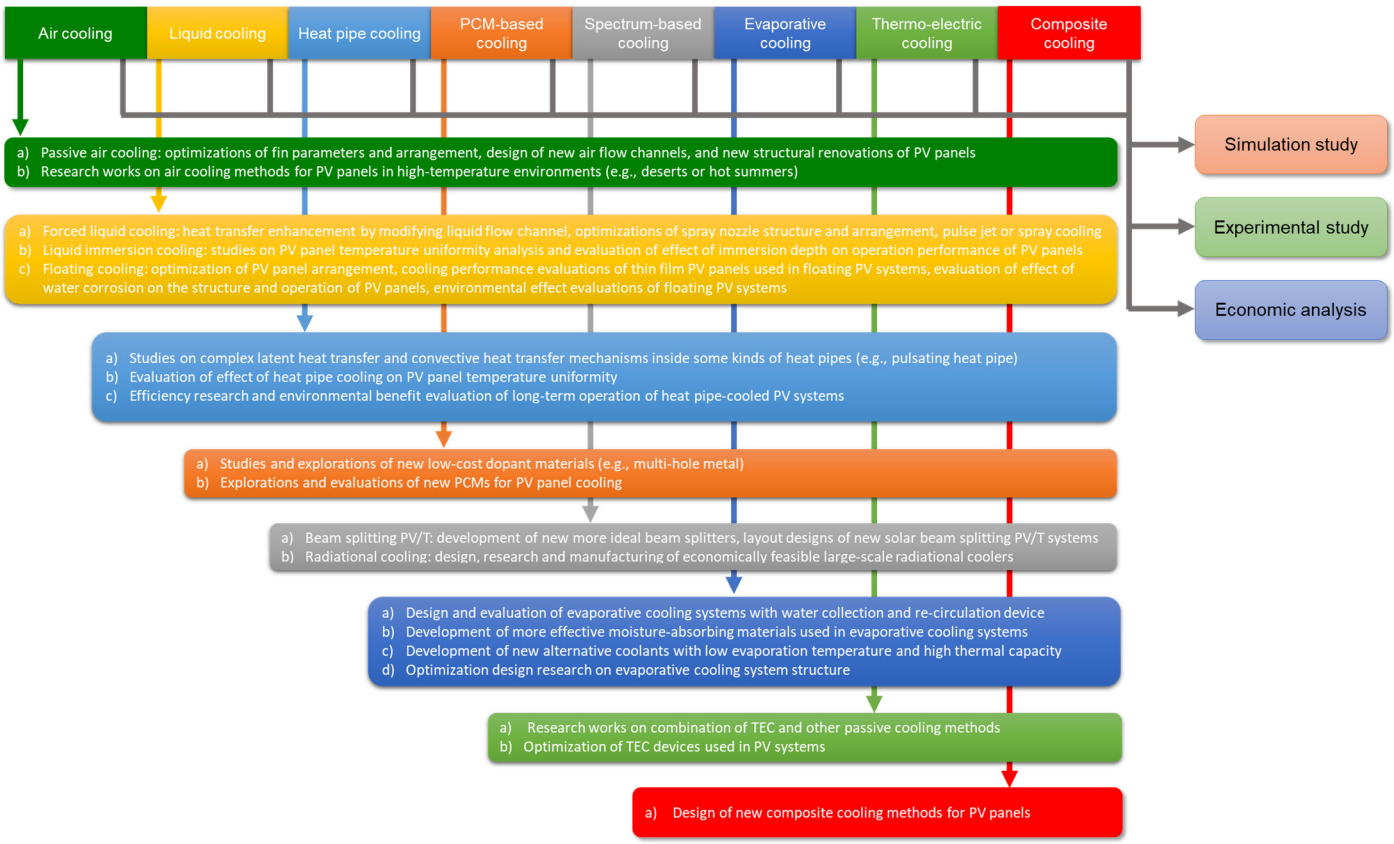


Figure 50. Brief roadmap of future research directions for different PV panel cooling methods.

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Abbreviations

BAPV	building-attached photovoltaic
CPV	concentrated solar photovoltaic
EVA	ethylene-vinyl acetate
O&M	operation and maintenance
PCM	phase-change material
PDMS	polydimethylsiloxane
PV	photovoltaic
PV/T	photovoltaic/thermal

SEM	scanning electron microscope
TEC	thermo-electric cooling

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